SMART MULTI-SENSOR WEARABLE FOR REAL-TIME ELDERLY HEALTH ANOMALY AND FALL DETECTION

INDIVIDUAL THESIS

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DECLARATION AND SUPERVISOR CERTIFICATION

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ABSTRACT

Stress is rarely measured in older adults except through occasional self-report. This thesis presents the Stress Monitoring System, my component of a multi-sensor wearable platform for elderly health monitoring. It has two parts. In Part A, a galvanic skin response (GSR) sensor on the glove's ESP32 is sampled at 50 Hz and smoothed. A hub-based engine then estimates stress with a Random Forest trained on statistical features of heart rate and finger temperature over 120 s windows, blends in a GSR score, and falls back to the vital-only model when the electrodes come off. In Part B, a photograph taken in the mobile application is passed through Firebase, the face is found with a Haar cascade, and a MobileNetV2 network fine-tuned by transfer learning labels it as stressed or not stressed. On 2,280 windows from 38 subjects held out of training, the physiological model reached an accuracy of 96.62%, a balanced accuracy of 96.22% and a stressed-class recall of 95.36%. The facial model, trained on 189 images, reached a best validation accuracy of 86.49% on 37 images and showed signs of overfitting during fine-tuning; its end-to-end processing took about 150 ms on a central processing unit (CPU). The results suggest that low-cost stress screening is technically feasible within the platform. However, the origin of the physiological dataset still has to be documented, and both models need testing on real data from older adults before anyone relies on them. My contribution covers the GSR acquisition on the glove, the hub-side stress engine and its evaluation, the facial stress pipeline, and their integration with the hub, Firebase, the dashboard and the mobile application.

Keywords: stress detection, galvanic skin response, random forest, MobileNetV2, older adults

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
ACC	Accelerometer
ACM	Association for Computing Machinery
ADC	Analog-to-Digital Converter
AI	Artificial Intelligence
BGR	Blue-Green-Red (colour channel order)
BVP	Blood Volume Pulse
CNN	Convolutional Neural Network
CPU	Central Processing Unit
CV	Cross-Validation
DNN	Deep Neural Network
ECG	Electrocardiogram
EDA	Electrodermal Activity
EMG	Electromyogram
FN	False Negative
FP	False Positive
GPIO	General-Purpose Input/Output
GPS	Global Positioning System
GSR	Galvanic Skin Response
HR	Heart Rate
HRV	Heart Rate Variability
HTTP	Hypertext Transfer Protocol
IEC	International Electrotechnical Commission
IEEE	Institute of Electrical and Electronics Engineers
IMU	Inertial Measurement Unit

Abbreviation	Meaning
IR	Infrared
ISO	International Organization for Standardization
JSON	JavaScript Object Notation
LOSO	Leave-One-Subject-Out
MDPI	Multidisciplinary Digital Publishing Institute
ML	Machine Learning
MQTT	Message Queuing Telemetry Transport
PIL	Python Imaging Library
RGB	Red-Green-Blue (colour channel order)
RQ	Research Question
SDG	Sustainable Development Goal
SDK	Software Development Kit
SLIIT	Sri Lanka Institute of Information Technology
SOS	Emergency distress signal
SSE	Server-Sent Events
SVM	Support Vector Machine
TN	True Negative
TP	True Positive
UDP	User Datagram Protocol
URL	Uniform Resource Locator
WESAD	Wearable Stress and Affect Detection (dataset)
YOLO	You Only Look Once

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1 INTRODUCTION

This chapter introduces the Stress Monitoring System, the individual component described in this thesis. It outlines the group project, explains where stress monitoring fits within it, and then states the problem, the research gap, the aim and objectives, the research questions, the scope and the individual contribution, before describing the structure of the thesis.

1.1 Overall Project Context

The group project, "Smart Multi-Sensor Wearable for Real-Time Elderly Health Anomaly and Fall Detection", developed a wearable monitoring system for older adults. In the project files the system is referred to by its working name, SmartGlove. The system consists of a smart glove built on an ESP32 microcontroller with a pulse oximeter, an infrared (IR) temperature sensor, an inertial measurement unit (IMU), a Global Positioning System (GPS) receiver, an electrocardiogram (ECG) module and a galvanic skin response (GSR) input; a smart belt that detects posture, impacts and falls; a local hub running a Flask backend that executes the machine learning (ML) models, sensor fusion and clinical rules; a Firebase Realtime Database; a web dashboard; and a mobile application. The hub combines the outputs of its screening engines into a Master Risk Score from 0 to 100, and an emergency alert is sent through Firebase when the score exceeds 60, when the belt detects a hard fall or when the glove detects an SOS shake gesture. Figure 1.1 shows the system and the position of the Stress Monitoring System within it. The other screening engines, the belt, the SOS gesture and the design of the Master Risk Score were developed by other members of the group and are outside the scope of this thesis.

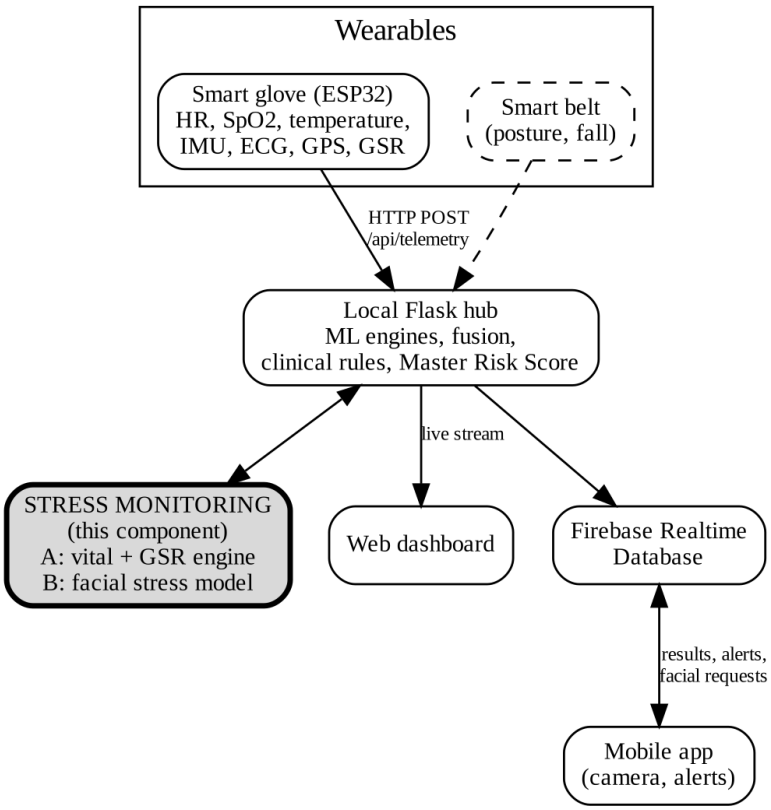


Figure 1.1: Overall system with the Stress Monitoring System highlighted

1.2 Individual Component Context

Psychological stress is the body's response to demands or threats it perceives. It engages the sympathetic nervous system and the hypothalamic-pituitary-adrenal axis, which show up as changes in heart rate, sweating, peripheral blood flow and facial expression [1], [2]. A short episode of stress is a normal, useful reaction, but when it is repeated or prolonged it is linked to harm to health [1]. Continuous, non-invasive monitoring is especially attractive for older adults, who may live alone, may find it hard or unappealing to say how they feel, and are more likely to have chronic conditions.

Within the group project, the Stress Monitoring System adds a psychological dimension to a platform that otherwise deals with cardiorespiratory, neurological and safety events. It has two parts, which are kept apart from each other and from the other members' components. Part A is a stress engine on the hub. It estimates the probability of a stress state from the heart rate and finger skin temperature measured by the glove, and blends in a score derived from the GSR signal. Part B is a facial stress detector: it

takes a photograph from the mobile application, finds the face, and classifies it as stressed or non-stressed with a fine-tuned convolutional neural network (CNN). Both parts give screening outputs for the user and carers. Neither is meant to diagnose a medical or psychiatric condition.

1.3 Problem Statement

Stress is usually assessed with questionnaires or interviews. These depend on self-report, happen only occasionally, and can be hard for some older adults to complete [3], [2]. Physiological signals such as electrodermal activity (EDA), heart rate and skin temperature change automatically under stress, so wearable sensors can follow them continuously [4], [5]. Facial expressions can be captured with an ordinary smartphone camera [6]. In practice, though, laboratory stress detectors often depend on research-grade devices, are evaluated in ways that let data from the same person appear in both training and test sets, and are rarely built into a complete monitoring system that older adults and their carers already use [7], [8].

The problem this thesis addresses is how to screen older adults for stress continuously and without intrusion, using the low-cost sensors already on a wearable glove and the camera of a mobile phone. The estimates also have to reach carers through the existing hub, dashboard and mobile application, and the models have to be evaluated in a way that reflects use on new people.

1.4 Research Gap

Chapter 2 shows that EDA, heart rate and skin temperature carry information about stress [4], [5], [9], and that deep learning can classify stress from facial images and video [6]. It also exposes four gaps that matter for this component. First, many reported accuracies come from record-wise evaluation, where windows from the same subject sit in both training and test data; this can overestimate performance on new users [7], [8]. Second, most wearable stress studies use research devices such as chest straps or dedicated wristbands, not low-cost sensors built into a multi-purpose wearable [3]. Third, physiological and facial approaches are usually studied separately, and few systems offer both on one platform. Finally, few studies target older adults or deliver stress estimates through a system that also handles other health alerts [8].

Table 1.1: Comparison of selected existing work with the proposed component

Work	Signals	Device	Evaluation	Integrated system
Healey and Picard [4]	ECG, EDA, EMG, respiration	Research sensors in a car	Real-world drives	No
Schmidt et al. [5]	EDA, BVP, temperature, ACC and others	Chest and wrist research devices	Leave-one-subject-out	No (dataset)
Gjoreski et al. [9]	BVP, EDA, temperature, ACC	Commercial research wristband	Laboratory and real life	No
Zhang et al. [6]	Facial video	Camera	Laboratory dataset	No
Proposed component	HR, finger temperature, GSR; face photo	Low-cost glove sensors; smartphone	Subject-wise split (Part A); hold-out (Part B)	Yes (hub, dashboard, app)

In short, what is missing is low-cost stress screening for older adults that is evaluated on unseen subjects, combines wearable physiological sensing with smartphone-based facial analysis, and fits into an existing health monitoring platform.

1.5 Aim and Objectives

1.5.1 Main aim

The main aim of this work was to design, implement and evaluate a Stress Monitoring System that screens for stress from wearable physiological signals and from facial images, and to integrate it into the group's elderly health monitoring platform.

1.5.2 Specific objectives

1. To acquire a GSR signal on the glove node with a stable sampling rate and noise filtering, and to transmit it to the hub together with the other vital signs.

2. To build a stress classifier on windowed heart-rate and skin-temperature features and to evaluate it with a subject-independent split, reporting accuracy, balanced accuracy, precision, recall and F1-score.

3. To combine the classifier output with a GSR score and to provide a fall-back when the GSR electrodes are detached.

4. To build a facial stress classifier using face detection and transfer learning on MobileNetV2, and to evaluate it on held-out images.

5. To integrate both parts with the Flask hub, Firebase, the web dashboard and the mobile application.

6. To measure the inference latency of both parts and to verify the integration with functional tests.

1.6 Research Questions or Hypotheses

The work addressed the following research questions:

RQ1. Can a RandomForest classifier trained on statistical features of heart rate and skin temperature over 120 s windows detect stress when evaluated on subjects not seen during training?

RQ2. Can a MobileNetV2 network, pre-trained on ImageNet and fine-tuned on a small set of facial images, classify stressed and non-stressed faces on held-out images?

RQ3. Can both stress estimates be delivered in near real time through the hub, Firebase, the dashboard and the mobile application?

1.7 Scope Limitations and Assumptions

1.7.1 Scope

The scope covers GSR acquisition on the glove node, the vital-sign and GSR stress engine on the hub, the facial stress detection pipeline, and their integration with the hub, Firebase, the web dashboard and the mobile application. The other screening engines (hypoxemia, arrhythmia, Parkinson's tremor and sepsis), fall detection, the SOS gesture, the Master Risk Score and the belt hardware belong to other members. They are mentioned only where needed to explain the integration.

1.7.2 Limitations

Section 7.5 discusses the main limitations in detail. In brief: the origin and labelling of the 150-subject stress dataset are not documented in the project files; the GSR score is a fixed-weight heuristic outside the trained model, with fixed analog-to-digital converter (ADC) bounds and no per-user baseline; the facial dataset is small (189 images), with a 37-image validation set and no separate test set; and no clinical validation with older adults was done.

1.7.3 Assumptions

It was assumed that the glove is worn with a finger on the pulse sensor so that telemetry is transmitted, that the GSR electrodes are in contact with two fingers, that the hub and Firebase are reachable, and that the simulated vital-sign mode of the glove (syncMode) is switched off whenever stress is evaluated.

1.8 Individual Contribution

The individual contribution of the author consisted of the following parts of the Stress Monitoring System, and their integration with the components of the other group members through the hub, Firebase, the web dashboard and the mobile application:

1. The GSR acquisition on the glove node: sampling, noise filtering and transmission to the hub together with the other vital signs (Objective 1).

2. The stress classifier on the hub: windowed heart-rate and skin-temperature features, the RandomForest model and its subject-wise evaluation (Objective 2).

3. The GSR score blend and the fall-back to the vital-only model when the electrodes are detached (Objective 3).

4. The facial stress pipeline: Base64 exchange through Firebase, Haar cascade face detection, the fine-tuned MobileNetV2 classifier and the /api/facial-stress endpoint (Objective 4).

5. The integration of both parts with the hub, Firebase, the web dashboard and the mobile application (Objective 5).

6. The measurement of inference latency and the functional verification of both parts (Objective 6).

1.9 Thesis Structure

Chapter 2 reviews the literature on stress physiology, wearable and facial stress detection, and the techniques used. Chapter 3 sets out the research methodology, datasets, evaluation metrics and ethical considerations. Chapter 4 presents the requirements and the design of both parts. Chapter 5 covers the implementation, the integration with the other components and the verification. Chapter 6 reports the

results and Chapter 7 discusses them. Chapter 8 looks at commercialisation and sustainability, and Chapter 9 concludes and suggests future work.

2 LITERATURE REVIEW AND RELATED WORK

This chapter reviews the literature that informed the Stress Monitoring System. It describes the review method, the physiological and computational foundations, related research on wearable and facial stress detection, existing products and techniques, and finally synthesises the research gap into a conceptual framework.

2.1 Review Method and Scope

A narrative literature review was carried out. Peer-reviewed journal articles, conference papers and books were searched in Institute of Electrical and Electronics Engineers (IEEE) Xplore, Association for Computing Machinery (ACM) Digital Library, PubMed, ScienceDirect, Multidisciplinary Digital Publishing Institute (MDPI) and Google Scholar using combinations of the terms listed in Table 2.1. Studies were included if they addressed stress detection from EDA, heart rate or skin temperature, facial stress or expression recognition, transfer learning or face detection methods used in this work, or the evaluation of ML models on physiological data. Studies limited to clinical anxiety disorders or using only questionnaires were excluded.

Table 2.1: Search terms and inclusion criteria

Aspect	Description
Search terms	"stress detection" AND (GSR OR "electrodermal activity" OR "heart rate" OR "skin temperature" OR wearable); "facial stress" OR "facial expression" AND stress; MobileNetV2 transfer learning; Haar cascade face detection; subject-independent OR "leave-one-subject-out"; stress AND "older adults"
Sources	IEEE Xplore, ACM Digital Library, PubMed, ScienceDirect, MDPI, Google Scholar
Included	Peer-reviewed studies, books and datasets on physiological or facial stress detection and the methods used
Excluded	Questionnaire-only studies; clinical diagnosis of anxiety disorders; non-peer-reviewed claims

2.2 Domain and Theoretical Foundations

2.2.1 Stress physiology

When a situation is perceived as threatening or demanding, the sympathetic nervous system prepares the body for action in what is commonly called the fight-or-flight response, and the hypothalamic-pituitary-adrenal axis releases cortisol [1]. These mediators are protective in the short term, but McEwen [1] described how their

repeated or prolonged activation produces an allostatic load that contributes to cardiovascular, metabolic and immune problems. Laboratory protocols such as the Trier Social Stress Test, which combines a public speaking task with mental arithmetic in front of an audience, are widely used to induce stress reliably for research [10], and the test was used to induce stress in the Wearable Stress and Affect Detection (WESAD) dataset [5].

2.2.2 Electrodermal activity

Eccrine sweat glands, which are dense on the palms and fingers, are innervated only by the sympathetic nervous system, so changes in their activity are a direct marker of sympathetic arousal [11]. EDA, also called GSR or skin conductance, is measured by passing a small current between two electrodes on the skin; as sweat fills the ducts, the skin's resistance falls and its conductance rises [11]. The signal has a slowly varying tonic level and faster phasic responses to stimuli. EDA differs greatly between individuals and is affected by temperature, humidity and electrode contact, which is why per-subject baselines or normalisation are commonly used [11], [2].

2.2.3 Heart rate and skin temperature

Stress usually increases heart rate and decreases heart rate variability (HRV), reflecting a shift of autonomic balance towards sympathetic activity; a meta-analysis by Kim et al. [12] concluded that low parasympathetic activity, measured by HRV, is associated with stress. Peripheral skin temperature also changes, because sympathetic vasoconstriction reduces blood flow to the fingers [2]. These signals are less specific than EDA, since physical activity, posture, ambient temperature and illness affect them as well, which is why they are usually combined with other signals and with context [9], [3].

2.2.4 Facial expression and stress

Stress and anxiety are associated with changes in facial expression, such as eye and mouth movements, head motion and blinking, and these cues can be captured by a camera without contact [2], [6]. Image-based methods do not require the user to wear sensors, but they capture only a moment, they are sensitive to lighting and pose, and posed expressions may differ from those produced by genuine stress.

2.2.5 Random forests

A random forest is an ensemble of decision trees, each trained on a bootstrap sample of the data and choosing each split from a random subset of features; the prediction is the average or majority vote of the trees [13]. Random forests handle non-linear relationships and feature interactions without feature scaling, are robust to noisy features, provide class probabilities and require relatively little tuning [13]. The implementation used in this work is that of the scikit-learn library [14].

2.2.6 Convolutional neural networks, transfer learning and MobileNetV2

CNNs learn hierarchical image features through stacked convolutional layers. Training a CNN from scratch requires large labelled datasets; transfer learning instead reuses a network pre-trained on a large dataset such as ImageNet and adapts it to a new task, which is effective when the target dataset is small [15]. A common procedure is to train a new classification head on top of the frozen pre-trained base, and then to fine-tune some of the upper layers of the base with a low learning rate.

MobileNetV2 [16] is a CNN designed for mobile and embedded devices. It is built from inverted residual blocks with linear bottlenecks and depthwise separable convolutions, which greatly reduce computation and the number of parameters compared with conventional CNNs while retaining good accuracy [16]. These properties make it suitable for a hub that must run several models on an ordinary computer.

2.2.7 Haar cascade face detection

Viola and Jones [17] proposed a face detector that computes simple rectangular Haar-like features quickly with an integral image, selects the most informative features with AdaBoost, and arranges classifiers in a cascade so that most non-face regions are rejected early. The method runs in real time on a central processing unit (CPU) and is available in OpenCV as pre-trained cascades such as haarcascade_frontalface_default. It works best on frontal faces and is sensitive to rotation and lighting.

2.2.8 Data augmentation

Data augmentation creates modified copies of training images, for example by flipping, rotating, zooming or changing brightness and contrast, so that the model

learns features that are invariant to these changes and overfits less on small datasets [18].

2.3 Related Research

2.3.1 Wearable physiological stress detection

In 2005, the study of "Detecting stress during real-world driving tasks using physiological sensors" [4] proposed recording ECG, EDA, electromyography and respiration from drivers on planned routes through city and highway traffic, and achieved classification of three stress levels, finding that skin conductivity and heart rate were the measures most closely correlated with driver stress. The study established EDA and heart rate as key signals for stress detection outside the laboratory.

In 2017, the study of "Monitoring stress with a wrist device using context" [9] proposed detecting stress from a commercial wristband measuring blood volume pulse, EDA, skin temperature and acceleration, using laboratory data to build the detector and adding context such as physical activity to reduce false alarms in real life. The study showed that context is important when stress is detected from signals that are also affected by activity.

In 2018, the study of "Introducing WESAD, a multimodal dataset for wearable stress and affect detection" [5] proposed a public dataset recorded from 15 subjects with chest-worn and wrist-worn devices during baseline, amusement and Trier Social Stress Test conditions, and achieved benchmark results using classical ML with leave-one-subject-out evaluation. WESAD is widely used as a reference for wearable stress detection.

Surveys have summarised this field. Can et al. [3] reviewed stress detection in daily life with smartphones and wearables and highlighted the gap between laboratory accuracy and real-life performance. Giannakakis et al. [2] reviewed the biosignals used for stress detection, including EDA, heart rate, skin temperature and facial cues. Vos et al. [8] reviewed ML for stress monitoring from wearable devices and found that many models generalise poorly to new subjects and datasets.

2.3.2 Subject-independent evaluation

In 2017, the study of "The need to approximate the use-case in clinical machine learning" [7] proposed comparing record-wise and subject-wise cross-validation on health data, and showed that record-wise validation, in which records from the same person appear in both training and test sets, can greatly overestimate the accuracy obtained on new people. For a stress monitor intended for new users, subject-wise evaluation is therefore the appropriate choice, and it is the reason the stress engine in this work was evaluated with a split grouped by subject.

2.3.3 Facial stress detection

In 2020, the study of "Video-based stress detection through deep learning" [6] proposed a deep learning framework that detects stress from facial video, combining facial expressions with action cues, and achieved results on a dataset collected for the purpose that showed facial video can be used for stress detection without contact sensors. Giannakakis et al. [2] also reported facial cues as one of the non-contact modalities for stress detection.

Table 2.2: Summary of related stress detection studies

Study	Signals	Method	Data	Relevance
[4]	ECG, EDA, EMG, respiration	Feature-based classification	Real-world driving	EDA and HR track stress
[9]	BVP, EDA, temperature, ACC	ML with activity context	Laboratory and real life	Context reduces false alarms
[5]	Chest and wrist signals	Classical ML, LOSO	WESAD, 15 subjects	Public benchmark
[7]	Health data (general)	Record-wise vs subject-wise CV	Several datasets	Subject-wise evaluation needed
[8]	Wearable signals	Systematic review	Published studies	Generalisation is weak
[6]	Facial video	Deep learning	Collected dataset	Non-contact facial stress

2.4 Existing Systems Products and Techniques

Several consumer devices offer stress-related features. The Fitbit Sense smartwatch introduced an EDA scan application that measures skin conductance while the user rests a palm on the watch face during a short guided session [19]. Many smartwatches also estimate stress from heart rate and HRV. These products are designed for healthy adults, rely on proprietary algorithms, and are not integrated with the other health alerts

needed by older adults and their carers. Research datasets and techniques, summarised in Table 2.3, provide the methods used in this work but not an integrated system.

Table 2.3: Comparison of existing products and techniques with the proposed component

Approach	Signals	Cost / availability	Open method	Integrated with elderly health alerts
Smartwatch EDA scan [19]	EDA (on demand)	Consumer device	No	No
Research wristbands [9]	BVP, EDA, temperature	Research device	Partly	No
Public datasets [5]	Multimodal	Free for research	Yes	No
Facial stress models [6]	Video / images	Camera	Yes (research)	No
Proposed component	HR, finger temperature, GSR; face photo	Low-cost glove sensors; phone	Yes	Yes

2.5 Gap Synthesis and Conceptual Framework

The review shows that EDA, heart rate and skin temperature respond to stress through the sympathetic nervous system [11], [12], that classical ML such as random forests can classify stress from windowed features [13], [5], and that facial images offer a contact-free alternative [6]. It also shows that evaluation must be subject-independent to reflect real use [7], [8], and that few systems integrate stress screening into a platform for older adults.

Figure 2.1 shows the conceptual framework used in this work. A stressor activates the sympathetic nervous system, which changes sweat gland activity, heart rate and finger temperature, and also changes facial expression. Part A measures the physiological changes with the glove and combines a trained classifier with a GSR heuristic. Part B looks at the facial changes in a photograph, using a face detector and a transfer-learned CNN. Both outputs are presented to the user and carers as non-diagnostic screening information.

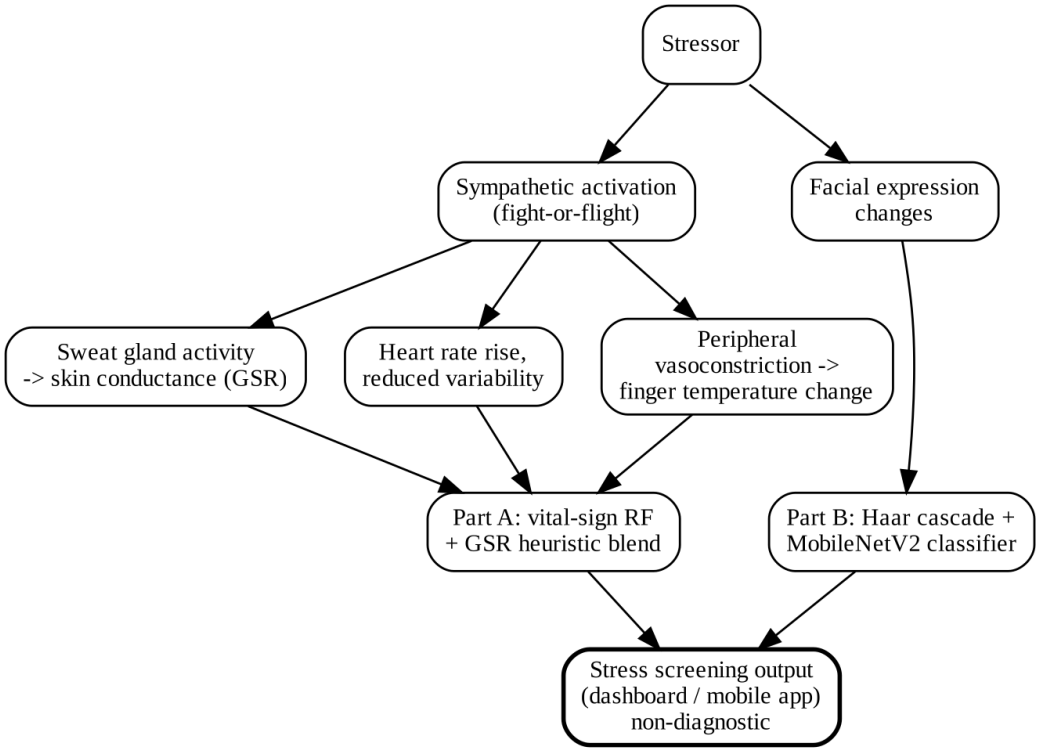


Figure 2.1: Conceptual framework of the Stress Monitoring System

3 RESEARCH METHODOLOGY

This chapter describes the research design, the data used for both parts of the component, the experimental procedure, the evaluation metrics, the data analysis, the ethical, privacy and security considerations, and the artefacts that support reproducibility.

3.1 Research Design

The research followed a design-and-build approach with quantitative evaluation. Two artefacts were developed: a wearable stress engine and a facial stress classifier. Each was evaluated with standard classification metrics on data not used for training, and both were integrated with the group's platform and checked with functional tests. This design was chosen because the research questions ask whether specific models and an integrated system can achieve a practical goal, which requires building and measuring them. The process followed is shown in Figure 3.1.

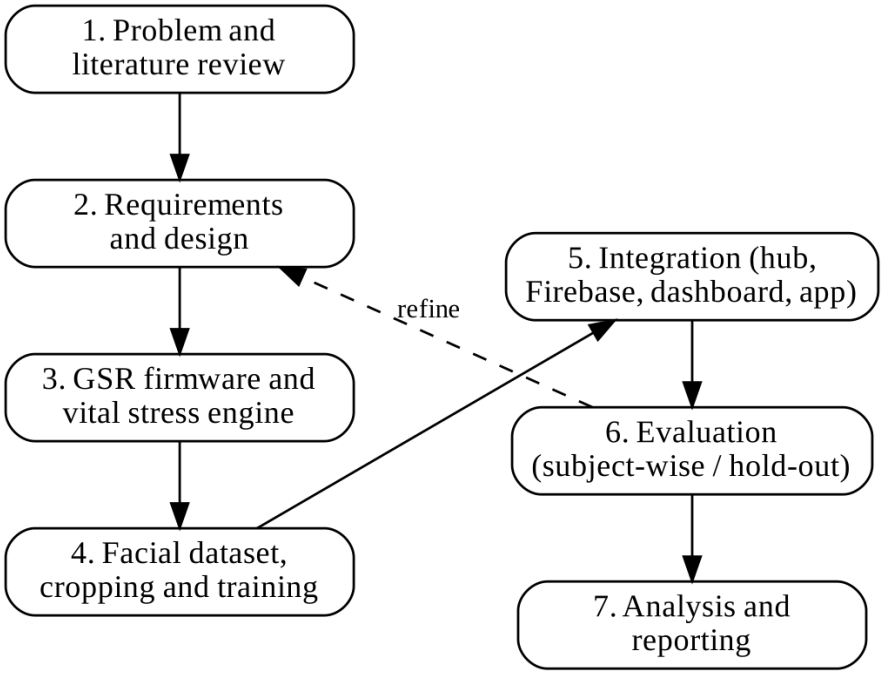


Figure 3.1: Research process followed for the Stress Monitoring System

3.2 Data Participants or Study Materials

3.2.1 Wearable stress dataset (Part A)

The stress engine was trained and evaluated on a dataset of 150 subjects and 9,000 windows, each window described by eight statistical features of heart rate and

temperature. The data were divided with a split grouped by subject into 112 training subjects (6,720 windows) and 38 test subjects (2,280 windows), so that no subject appears in both sets. The test set contains 1,677 not-stressed windows (73.5%) and 603 stressed windows (26.5%). The metrics file of the trained model records the sensor calibration as "MAX30102 + MLX90614 (SmartGlove)".

3.2.2 Facial stress dataset (Part B)

The facial dataset contains 189 images, of which 96 are labelled stressed and 93 non-stressed. The images were cropped to the face with a Haar cascade before training (Section 4.4.3) and stored in two class folders. They were split 80/20 into training and validation subsets by the Keras utility image_dataset_from_directory with a fixed seed, which for 189 images gives 152 training and 37 validation images. Figure 3.2 shows part of the image folder; the faces have been blurred to protect the identity of the people shown.

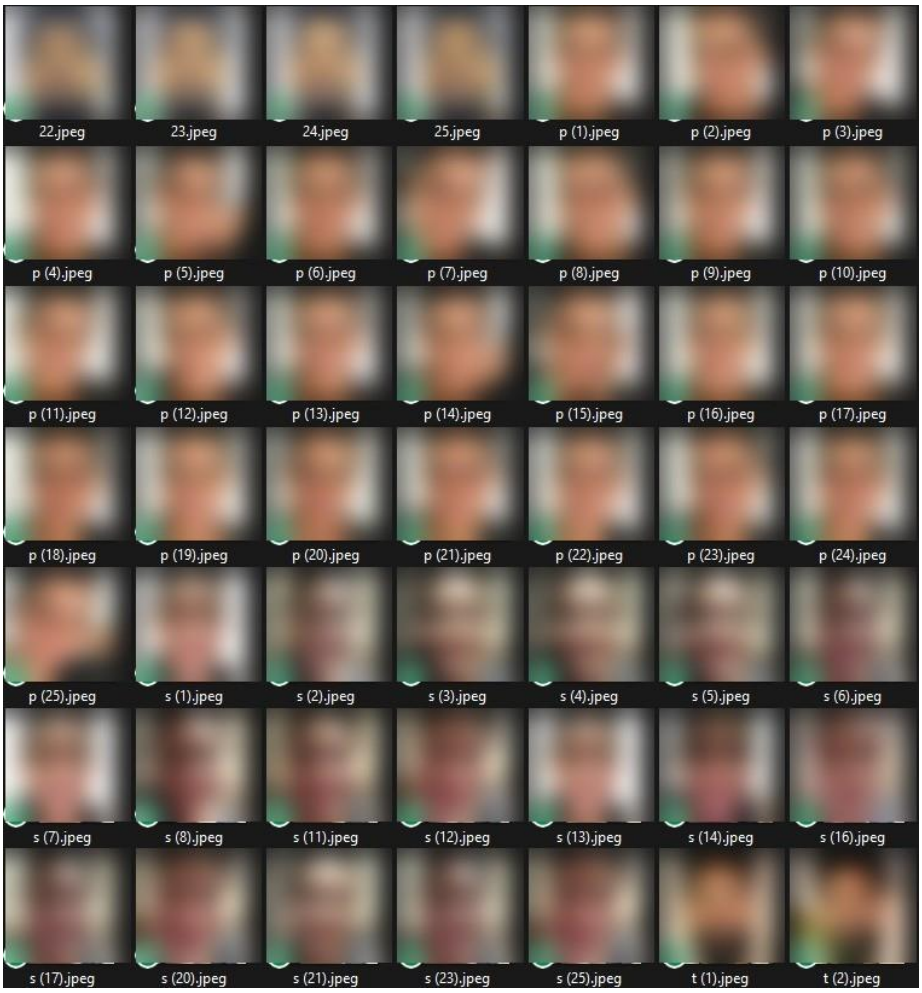


Figure 3.2: Part of the facial image dataset folder, faces blurred for privacy (Original in Color)

The file names in the folder use a small number of prefixes, which suggests that the images were taken from a small number of individuals, and the people shown do not appear to be older adults.

3.2.3 Hardware and software materials

The hardware consisted of the glove node (ESP32 Dev Module, WROOM-32) with an analog GSR module wired to general-purpose input/output (GPIO) 34, together with the MAX30100 pulse oximeter and MLX90614 IR temperature sensor already present on the glove. The software consisted of the glove firmware (glove.ino), a standalone GSR test sketch (test_gsr.ino), the Flask hub, the facial model training script (train_model.py), the inference class (facial_keras.py) and the trained model file (stress_classifier_finetuned.keras).

3.3 Experimental or Study Procedure

3.3.1 Part A procedure

The stress engine was trained on the 112 training subjects and evaluated once on the 38 held-out subjects, producing the metrics file stress_model.metrics.json. In addition, a test script (test_stress.py) was prepared that applies the trained model to eight hand-crafted feature vectors representing relaxed, mildly stressed, stressed and exercise conditions, and reports whether each is classified as expected.

3.3.2 Part B procedure

The facial model was trained in two phases with train_model.py. In the first phase the MobileNetV2 base was frozen and only the new classification head was trained for 15 epochs with the Adam optimiser at a learning rate of 1e-3. In the second phase, the layers of the base from index 100 onwards were unfrozen and the whole model was trained further with RMSprop at a learning rate of 1e-5, for up to 20 additional epochs, with early stopping on validation accuracy (patience 5, restoring the best weights). The final model was evaluated on the 37 validation images, and the accuracy and loss curves of both phases were plotted.

For this thesis, the inference time of the trained model was measured by loading stress_classifier_finetuned.keras with Keras 3.15.1 and timing 50 predictions after five warm-up runs, and the complete pipeline of facial_keras.py (Base64 decoding, Haar cascade detection, cropping and prediction) was timed on 48 face images. The measurements were made on a 2-vCPU Intel Xeon processor at 2.10 GHz, not on the hub computer.

3.4 Measures and Evaluation Metrics

Both parts solve a binary classification problem in which the positive class is "stressed". For each evaluated sample, a true positive (TP) is a stressed sample predicted as stressed, a false negative (FN) is a stressed sample predicted as not stressed, a true negative (TN) is a not-stressed sample predicted as not stressed, and a false positive (FP) is a not-stressed sample predicted as stressed. The metrics in Table 3.1 were calculated from these counts.

Table 3.1: Evaluation metrics

Metric	Formula	Use
Accuracy	(TP + TN) / (TP + TN + FP + FN)	Parts A and B
Precision (stressed)	TP / (TP + FP)	Parts A and B
Recall / sensitivity (stressed)	TP / (TP + FN)	Parts A and B
Specificity	TN / (TN + FP)	Part A
F1-score	2 × Precision × Recall / (Precision + Recall)	Per class
Balanced accuracy	(Recall of stressed + Specificity) / 2	Part A (imbalanced test set)
Macro F1-score	Mean of the per-class F1-scores	Parts A and B
Binary cross-entropy loss	$-\text{mean}[y \log p + (1 - y) \log(1 - p)]$	Part B training curves
Inference latency	Time per prediction (ms)	Parts A and B

Balanced accuracy was reported for Part A because the test set is imbalanced (73.5% not stressed), so plain accuracy would favour the majority class. For Part B, the evaluation code computes precision and recall for the positive (stressed) class, while the F1-score is macro-averaged over the two classes; this difference is stated wherever these values are reported.

3.5 Data Analysis

For Part A, the classification report and confusion matrix produced by scikit-learn [14] on the 2,280 test windows were taken from the metrics file and checked by recomputing accuracy, balanced accuracy, precision, recall and F1-score from the confusion matrix. For Part B, the training history was analysed from the accuracy and loss curves, and the validation accuracy values were related to the number of validation images (37), since every possible accuracy value is a multiple of 1/37. The precision, recall, macro F1-score and confusion matrix of the facial model must be obtained by running the evaluation code on the validation set.

3.6 Ethics Privacy Security and Responsible AI

This section covers the ethical, privacy and security aspects of the Stress Monitoring System, including the responsible use of artificial intelligence (AI) in its two models.

3.6.1 Informed consent

Facial images are personal data that directly identify a person, and physiological data reveal information about health. Every person whose face or signals were used should

give informed written consent that covers collection, use for model training and publication. Older adults, who are the target users, may include people with reduced capacity to consent, for whom additional safeguards are required.

3.6.2 Privacy and data protection

Under Sri Lanka's Personal Data Protection Act, No. 9 of 2022, health data and biometric data are treated as special categories of personal data that require a lawful basis, purpose limitation, security safeguards and limited retention [20]. In the current design, the mobile application writes the full Base64 image to Firebase and the backend writes the result back. To protect privacy, images should be deleted from Firebase as soon as the result is produced, access to the facial_stress paths should be restricted to authenticated users by Firebase security rules, and only the classification result should be retained. Identifiable faces are not reproduced in this thesis (Figure 3.2 is blurred).

3.6.3 Security

The glove sends telemetry to the hub over plain Hypertext Transfer Protocol (HTTP) on the local network, and its source code contains the Wi-Fi credentials and the Firebase project address; these are not reproduced in this thesis. The hub discovery request to Firebase disables certificate verification. These weaknesses mean that telemetry could be read or altered by a device on the same network, and they are recorded as limitations in Section 7.5.

3.6.4 Bias and responsible AI

Both models may be biased. The facial dataset contains few individuals, apparently of a narrow age range, and possibly posed expressions, so the model may perform differently for older adults, for other skin tones, with glasses or in poor lighting. The physiological model was trained on a dataset whose population is undocumented. Both outputs are therefore presented as screening indicators, not diagnoses: the system indicates that a stress pattern may be present and does not diagnose stress, anxiety or any medical condition. Decisions about care remain with carers and clinicians.

3.7 Reproducibility and Research Artefacts

The research artefacts are the glove firmware (glove.ino), the GSR test sketch (test_gsr.ino), the stress model bundle and its metrics file (stress_model.joblib and stress_model.metrics.json), the stress test script (test_stress.py), the facial training script (train_model.py), the inference class (facial_keras.py), the trained facial model (stress_classifier_finetuned.keras, saved with Keras 3.15.1 on 6 October 2026) and the training curves. Fixed random seeds were used for the facial data split (seed 123) and the RandomForest (random_state 42). Table 3.2 lists the main configuration values.

Table 3.2: Main configuration values of the Stress Monitoring System

Item	Value
GSR input	GPIO 34 (ADC1_CH6), 12-bit, 11 dB attenuation
GSR sampling and filter	50 Hz (20 ms), 20-sample moving average
Stress features	Mean, std, min, max of heart rate and temperature over 120 s
RandomForest	300 trees, min_samples_leaf 2, max_features sqrt, class_weight balanced, random_state 42, median imputer
GSR blend	P_final = 0.8 P_model + 0.2 S_gsr; S_gsr from ADC 300-3200
Facial input	224 × 224 red-green-blue (RGB) face crop, 10% padding
Facial training	Batch 16; head: Adam 1e-3, 15 epochs; fine-tuning: RMSprop 1e-5, from layer 100, up to 20 epochs, early stopping patience 5
Facial data split	80/20 (152 / 37 images), seed 123

4 PRODUCT AND SOLUTION DESIGN

This chapter presents the design of the Stress Monitoring System: its requirements, its place in the architecture of the complete solution, the design alternatives considered, and the detailed design of the GSR front end, the vital-sign stress engine and the facial stress pipeline.

4.1 Requirements

4.1.1 Functional requirements

Table 4.1 lists the functional requirements, derived from the problem statement, the literature and the interfaces of the group's platform.

Table 4.1: Functional requirements

ID	Requirement
FR1	The glove shall sample the GSR signal, filter it and include it in every telemetry packet as gsr_raw.
FR2	The hub shall buffer heart rate and temperature and compute windowed statistical features.
FR3	The hub shall estimate the probability of a stress state with a trained classifier.
FR4	The hub shall blend the classifier output with a GSR score when valid GSR data are available, and fall back to the vital-only model otherwise.
FR5	The hub shall present the stress estimate on the web dashboard.
FR6	The mobile application shall send a face photograph for analysis through Firebase.
FR7	The hub shall detect the face, classify it as stressed or normal and return the class, confidence and bounding box.
FR8	The hub shall report clearly when no face is detected or the model is unavailable.
FR9	The facial analysis shall also be available through an HTTP endpoint (POST /api/facial-stress).

4.1.2 Non-functional requirements

Table 4.2: Non-functional requirements

ID	Quality	Requirement
NFR1	Generalisation	The physiological model shall be evaluated on subjects not used for training.
NFR2	Latency	Each stress estimate shall be produced fast enough for near-real-time display (well under one second per prediction).
NFR3	Robustness	Detached GSR electrodes or missing values shall not stop the stress estimate.
NFR4	Low cost	The component shall use low-cost sensors and an ordinary computer as the hub.
NFR5	Privacy	Facial images and physiological data shall be protected and kept only as long as needed.
NFR6	Transparency	Outputs shall be presented as non-diagnostic screening information with a confidence value.

4.2 Component Architecture within the Complete Solution

Figure 4.1 shows the architecture of the component and its interfaces. In Part A, the GSR module and the existing heart-rate and temperature sensors are read by the glove's ESP32, which sends one JavaScript Object Notation (JSON) packet per second to the hub's /api/telemetry endpoint by HTTP POST over Wi-Fi while a finger is detected on the pulse sensor. On the hub, the stress engine updates a rolling buffer, computes the features, runs the RandomForest and blends the result with the GSR score; the estimate is displayed on the web dashboard. In Part B, the mobile application captures a photograph and writes it as a Base64 string to Firebase at /facial_stress/request; a listener on the hub passes it to the facial stress detector, which writes the result to /facial_stress/result, where the mobile application reads it. The same detector can be called directly through POST /api/facial-stress.

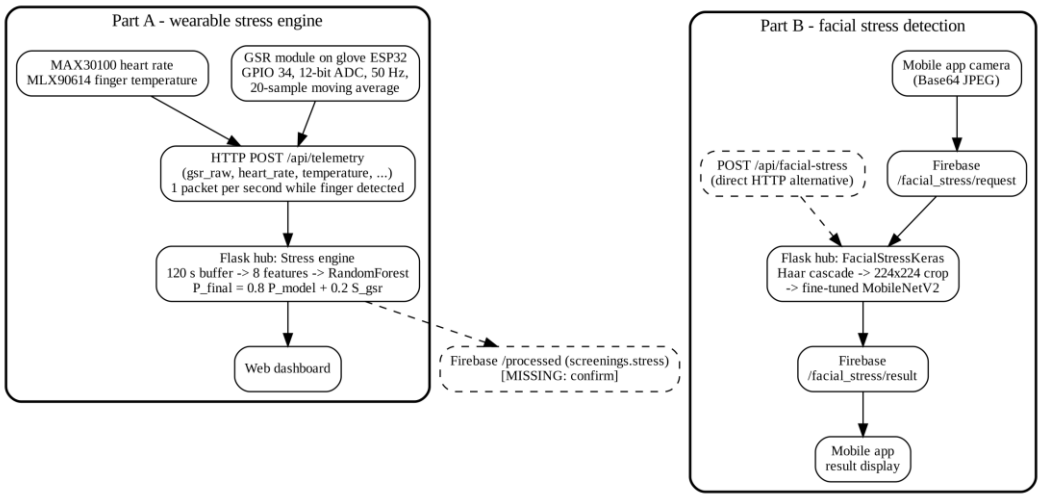


Figure 4.1: Architecture of the Stress Monitoring System and its interfaces

The other components interact with the stress component only through the hub. The glove firmware, shared with other members, carries the GSR value in the same packet as the other vital signs. The hub's other engines, the Master Risk Score and the alerting logic are owned by other members; whether the stress estimate contributes to the Master Risk Score is discussed in Section 5.3.

4.3 Design Alternatives and Decisions

4.3.1 Classifier for physiological signals

Classical ML on windowed statistical features was chosen over deep learning on raw signals. A random forest provides class probabilities, handles non-linear relationships between heart rate and temperature, needs no feature scaling and runs in a few milliseconds on a CPU [13], which matches the hub's requirement to run several models together. Deep networks would need much more labelled data and computation.

4.3.2 GSR as a heuristic blend rather than a model feature

The GSR value was not included as a feature of the trained model. Instead it is converted to a score between 0 and 1 and blended with the model probability with a fixed weight of 0.2. According to the project's engineering guide, this was done because raw GSR varies with humidity and skin moisture and because the training dataset did not contain a compatible GSR signal. The blend allows GSR to raise or lower the estimate without retraining, and allows the engine to fall back to the vital-only model when the electrodes are detached. Its weakness, discussed in Section 7.5, is that the weight and bounds were fixed rather than learned or validated.

4.3.3 Hardware moving average for GSR

A 20-sample moving average at 50 Hz was chosen to smooth the GSR signal. EDA changes over seconds [11], so a 0.4 s average removes electrical noise without hiding meaningful changes, and it requires only one addition and one subtraction per sample on the microcontroller.

4.3.4 Face detector

The OpenCV Haar cascade [17] was chosen for face detection because it runs quickly on a CPU, requires no additional model download and is reliable for the frontal, close-range photographs taken with a phone. Its weakness with rotated faces and poor lighting is accepted for this use.

4.3.5 Facial classifier

MobileNetV2 [16] with ImageNet weights was chosen because transfer learning is the most practical approach with only 189 images [15], and because MobileNetV2 is small

and fast enough to run on the hub alongside the other models. A two-phase strategy, training the head first and then fine-tuning the upper layers with a very low learning rate, was used to avoid destroying the pre-trained features. Project documents also describe a YOLOv8-based facial detector, but the trained model provided and the inference class facial_keras.py use MobileNetV2, which is the version described in this thesis.

Table 4.3: Summary of design decisions

Decision	Alternatives	Choice	Reason
Physiological classifier	Deep learning on raw signals; SVM	RandomForest on 8 window features	Small data, CPU speed, probabilities
Use of GSR	GSR as a model feature	Fixed-weight blend (0.2) with fall-back	No GSR in training data; robustness to detachment
GSR filtering	Software low-pass filter	20-sample moving average at 50 Hz	Simple, sufficient for slow EDA
Face detector	DNN or YOLO detectors	Haar cascade	Fast on CPU, adequate for frontal photos
Facial classifier	Training from scratch; YOLOv8	MobileNetV2 transfer learning	Small dataset, small fast model

4.4 Detailed Design

4.4.1 GSR front end

Figure 4.2 shows the GSR signal chain. The analog GSR module (a generic module with VCC, GND and SIG pins; the exact model is not known) is connected to two finger electrodes, on the index and middle fingers, and its output is read on GPIO 34 of the ESP32, which is channel 6 of ADC1. ADC1 was chosen because ADC2 cannot be used while Wi-Fi is active on the ESP32. The ADC is set to 12-bit resolution (0-4095) and 11 dB attenuation, which gives an input range of approximately 0-3.3 V. The signal is sampled every 20 ms (50 Hz) and smoothed with a 20-sample moving average, and the averaged value is sent as gsr_raw in raw ADC counts; it is not converted to microsiemens.

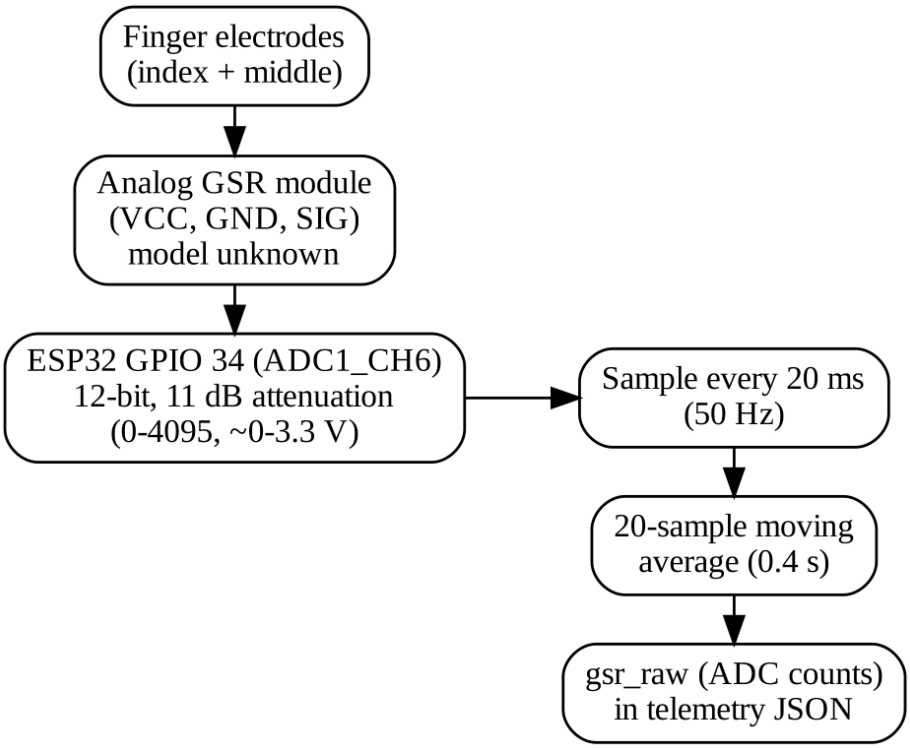


Figure 4.2: GSR signal chain on the glove node

A standalone test sketch, test_gsr.ino, was used to check the sensor. It prints the raw and filtered values, the voltage and an estimated skin resistance assuming a 100 kΩ voltage divider, and assigns one of three states using fixed thresholds on the filtered value, as listed in Table 4.4. No per-user baseline or calibration is implemented in either sketch. The project's developer guide also describes typical bands of about 800 to 2200 at rest, above 2800 for arousal and above 3500 for acute stress; these are descriptive only, and the stress engine itself uses the 300 to 3200 normalisation bounds described in Section 4.4.2.

Table 4.4: GSR states used in the test sketch

Filtered ADC value	State
Below 500	Electrodes detached
500 to 2800	Normal / relaxed
Above 2800	High conductance / stress

4.4.2 Vital-sign stress engine

Figure 4.3 shows the design of the stress engine on the hub. For each incoming packet, heart rate and temperature are appended to a rolling buffer covering 120 s. After a

warm-up of 10 samples, eight features are computed: the mean, standard deviation, minimum and maximum of heart rate, and the same four statistics of temperature. Statistical moments over a window were chosen rather than instantaneous values so that short artefacts, such as a movement of the finger, have less influence. The features are passed to a scikit-learn pipeline consisting of a median imputer, which replaces missing values with the training median, and a RandomForest with 300 trees, a minimum of two samples per leaf, the square root of the number of features considered at each split, balanced class weights and random_state 42. The output is the probability of the stressed class, P_model.

The GSR score is computed from the mean ADC value of the GSR samples in the buffer by linear scaling between fixed bounds and clipping:

$$S_{gsr} = \text{clip}((\text{mean ADC} - 300) / (3200 - 300), 0, 1) \quad (4.1)$$

and the final probability is

$$P_{final} = 0.8 \times P_{model} + 0.2 \times S_{gsr} \quad (4.2)$$

If the electrodes are detached or fewer than 10 valid GSR samples are available, the engine uses P_final = P_model. The design follows the project's engineering guide.

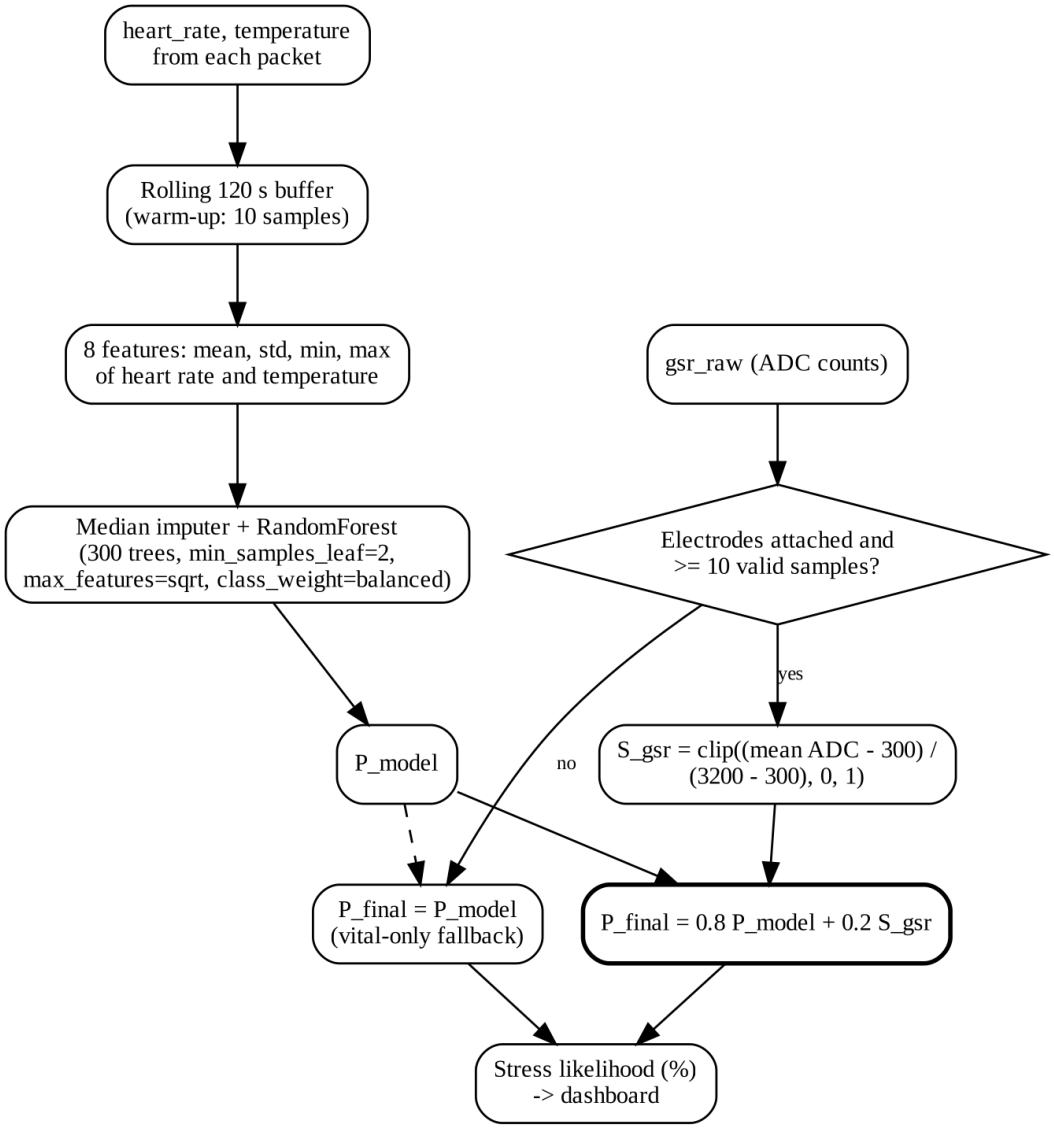


Figure 4.3: Design of the vital-sign stress engine with GSR blending

4.4.3 Facial stress pipeline

Figure 4.4 shows the inference pipeline implemented in the FacialStressKeras class. An image received as a Base64 string has any data uniform resource locator (URL) prefix removed and its padding corrected before decoding; it is opened with the Python Imaging Library (PIL) as an RGB image and converted to OpenCV's blue-green-red (BGR) format and then to greyscale. The Haar cascade detector is run with a scale factor of 1.1, a minimum of five neighbours and a minimum face size of 30 × 30 pixels. If no face is found, the result status is no_face_detected. Otherwise the largest face is selected, its bounding box is enlarged by 10% on each side, clipped to the image,

cropped and resized to 224 × 224 pixels, and converted back to RGB. The same detection, padding and resizing were applied to the training images by the cropping script (crop_faces.py), so that training and inference inputs are consistent.

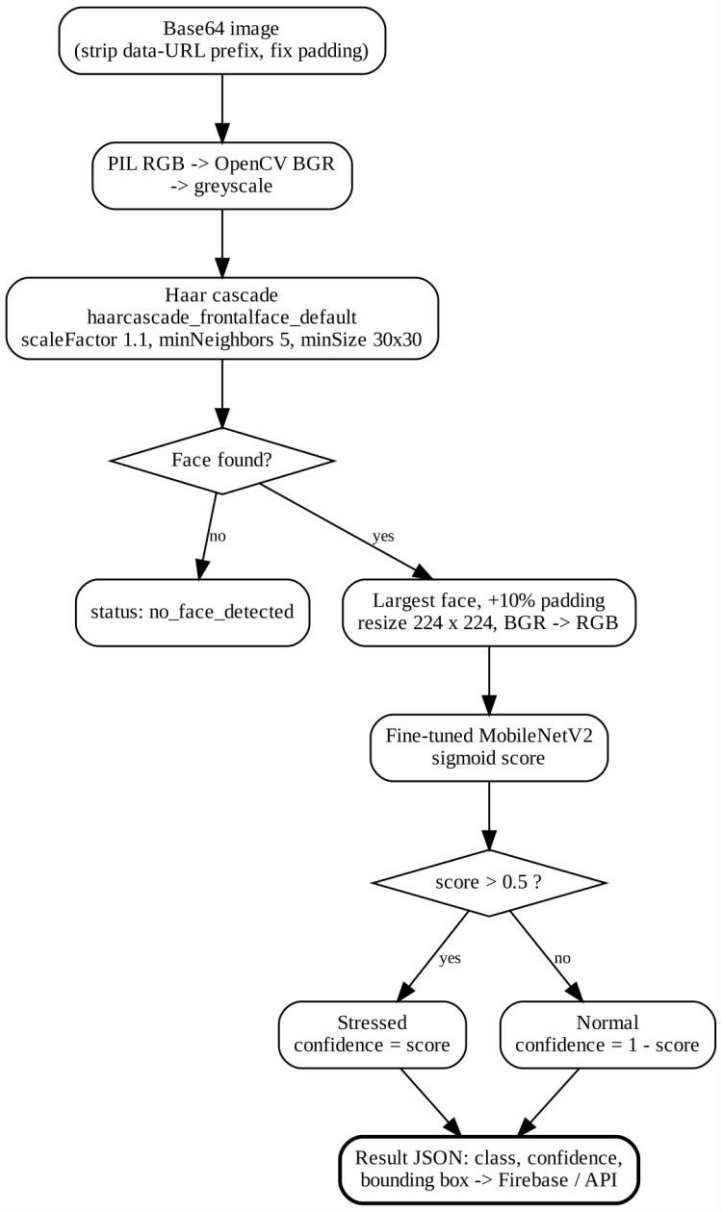


Figure 4.4: Facial stress inference pipeline

Figure 4.5 shows the network. The input passes through a data augmentation block that is active only during training (random horizontal flip, rotation of up to 0.1 of a full turn, zoom of up to 10%, contrast and brightness changes of up to 20%), through the MobileNetV2 preprocessing function, which scales pixel values to the range −1 to 1,

and through the MobileNetV2 base without its original classification layers. A global average pooling layer, a dropout layer with rate 0.3 and a single sigmoid output unit form the new head. The output is the probability that the face is stressed; a score above 0.5 is classified as stressed, and the reported confidence is the score for stressed predictions and one minus the score for normal predictions. Inspection of the saved model showed that the MobileNetV2 base has 154 layers, of which the 54 layers from index 100 onwards were trainable during fine-tuning, and that the model has 2,259,265 parameters in total.

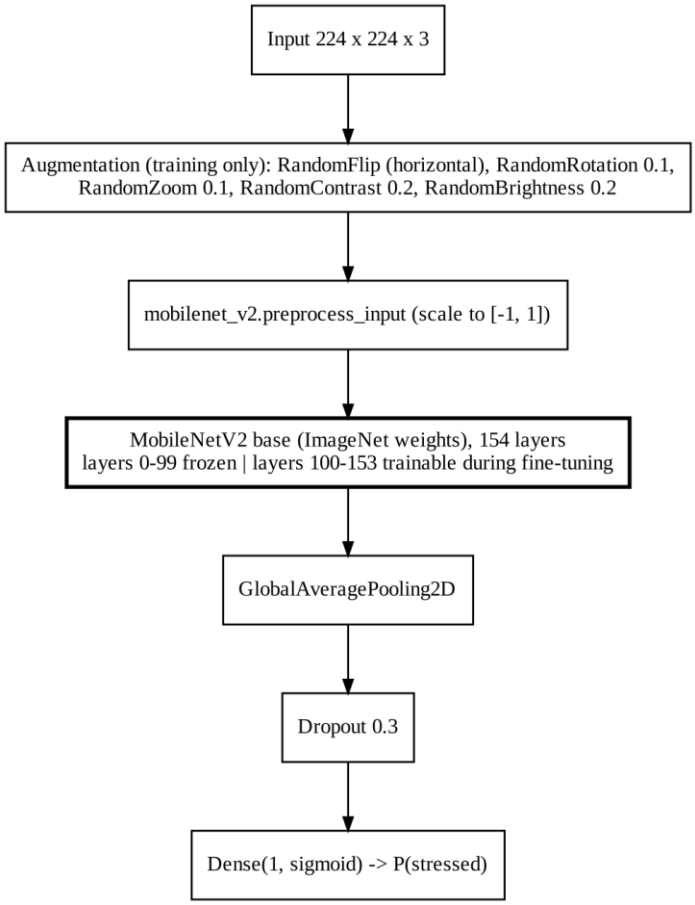


Figure 4.5: Structure of the fine-tuned MobileNetV2 stress classifier

4.5 Standards and Professional Practice

The component uses standard interfaces: HTTP and JavaScript Object Notation (JSON) for telemetry and the facial endpoint, and the Firebase Realtime Database for exchange with the mobile application. The ML work followed established practice by separating training and evaluation data by subject for Part A [7], fixing random seeds,

saving models with their metrics, and reporting balanced accuracy for an imbalanced test set.

A stress screening function is not intended as a medical device in its current form. If it were developed into a product making health claims, medical device software standards such as IEC 62304 [21] and risk management according to ISO 14971 [22] would apply; no claim of compliance is made. Personal data were considered in line with Sri Lanka's Personal Data Protection Act [20].

5 IMPLEMENTATION INTEGRATION AND TESTING

This chapter describes the development environment, the implementation of each part of the component with short code excerpts, the integration with the other components, the verification activities and the deployment. Wi-Fi credentials and Firebase addresses have been replaced by placeholders in all excerpts.

5.1 Development Environment and Technology Stack

Table 5.1 lists the technology stack and the main hyperparameters of the component.

Table 5.1: Technology stack and hyperparameters

Item	Details
Glove firmware	ESP32 Dev Module (WROOM-32), Arduino C++; libraries: WiFi, WiFiUdp, HTTPClient, WiFiClientSecure, Wire, MAX30100_PulseOximeter, Adafruit_MLX90614, Adafruit_MPU6050, Adafruit_SSD1306, TinyGPSPlus
Hub backend	Python 3.x, Flask (>= 3.0.0), Firebase Admin SDK (>= 6.5.0)
Machine learning	scikit-learn (>= 1.3.0), joblib, NumPy, Pandas
Deep learning	TensorFlow / Keras (model saved with Keras 3.15.1)
Image processing	OpenCV (opencv-python 4.10.0.84), Pillow
Stress engine	Median imputer + RandomForest (300 trees, min_samples_leaf 2, max_features sqrt, class_weight balanced, random_state 42); GSR weight 0.2
Facial model	MobileNetV2 (ImageNet), input 224 × 224 × 3, GlobalAveragePooling2D, Dropout 0.3, Dense(1, sigmoid)
Facial training	Batch size 16; head: Adam, lr 1e-3, 15 epochs; fine-tuning: RMSprop, lr 1e-5, from layer 100, up to 20 epochs; EarlyStopping (val_accuracy, patience 5, restore_best_weights)

5.2 Component Implementation

5.2.1 GSR acquisition firmware

In the glove firmware, the ADC is configured once in setup(), and the GSR is sampled in the main loop every 20 ms. A running total is kept so that the moving average is updated in constant time: the oldest sample is subtracted, the new sample is stored and added, and the total is divided by 20. The averaged value is written into the telemetry JSON as gsr_raw, as shown in Figure 5.1.

```
analogReadResolution(12);           // 0-4095
analogSetAttenuation(ADC_11db);     // about 0-3.3 V
...
if (millis() - lastGsrSampleTime >= 20) {          // 50 Hz
  lastGsrSampleTime = millis();
  int rawGSR = analogRead(GSR_PIN);                 // GPIO 34
  gsrTotal -= gsrReadings[gsrReadIndex];
  gsrReadings[gsrReadIndex] = rawGSR;
  gsrTotal += rawGSR;
  gsrReadIndex = (gsrReadIndex + 1) % NUM_GSR_READINGS;   // 20 samples
```

```
  averageGSR = gsrTotal / NUM_GSR_READINGS;
}
...
json += "\"gsr_raw\":" + String(averageGSR) + ",";
```

Figure 5.1: GSR sampling, moving average and telemetry field (excerpt from glove.ino)

Two properties of the glove firmware affect the stress engine. First, telemetry is sent only when a finger is detected on the MAX30100 sensor (raw infrared value above 34,000), when an SOS is active, or when syncMode is on; without a finger on the pulse sensor, no stress data reach the hub. Second, when syncMode is switched on from the hub's settings, the firmware replaces heart rate, SpO2 and temperature with simulated healthy values (around 72.5 bpm, 98-99% and 36.65 °C), while gsr_raw is never simulated. Because the stress model uses heart rate and temperature, any data recorded with syncMode on are invalid for stress evaluation; the telemetry packet includes a sync_mode flag that allows such data to be identified.

5.2.2 Vital-sign stress engine

Figure 5.2 summarises the logic of the stress engine as documented in the project's engineering guide.

```
# Pipeline (trained offline)
pipeline = Pipeline([
    ("imputer", SimpleImputer(strategy="median")),
    ("classifier", RandomForestClassifier(n_estimators=300, min_samples_leaf=2,
        max_features="sqrt", class_weight="balanced", random_state=42))])

# Inference on the hub (per update)
p_model = pipeline.predict_proba(features_8)[0][1]          # P(stressed)
if gsr_attached and n_valid_gsr >= 10:
    s_gsr = np.clip((np.mean(gsr_adc) - 300) / (3200 - 300), 0.0, 1.0)
    p_final = 0.8 * p_model + 0.2 * s_gsr
else:
    p_final = p_model                                         # vital-only
fallback
```

Figure 5.2: Stress engine pipeline and GSR blend (reconstructed from the engineering guide)

5.2.3 Face cropping for the training data

The script crop_faces.py applied the Haar cascade to every original image and saved the padded, resized face crops into the dataset_cropped folder used for training. The core of the cropping logic, which is the same as in the inference class, is shown in Figure 5.3. Images in which no face was detected were excluded.

```
gray = cv2.cvtColor(cv_img, cv2.COLOR_BGR2GRAY)
faces = face_cascade.detectMultiScale(gray, scaleFactor=1.1,
                                      minNeighbors=5, minSize=(30, 30))
faces = sorted(faces, key=lambda f: f[2] * f[3], reverse=True)   # largest face
x, y, w, h = faces[0]
```

```
pad_w, pad_h = int(w * 0.1), int(h * 0.1)                          # 10% padding
x_pad, y_pad = max(0, x - pad_w), max(0, y - pad_h)
w_pad = min(cv_img.shape[1] - x_pad, w + 2 * pad_w)
h_pad = min(cv_img.shape[0] - y_pad, h + 2 * pad_h)
face = cv2.resize(cv_img[y_pad:y_pad + h_pad, x_pad:x_pad + w_pad], (224, 224))
```

Figure 5.3: Face detection, padding and resizing (excerpt from crop_faces.py)

5.2.4 Model training

Figure 5.4 shows the essential parts of train_model.py. The dataset is loaded from the class folders with an 80/20 split and seed 123, cached and prefetched. The model is built as described in Section 4.4.3 and trained in two phases. In the second phase, the base is unfrozen, the first 100 layers are frozen again, and the model is recompiled with RMSprop at 1e-5 before training continues from the last head epoch with early stopping.

```
train_ds = keras.utils.image_dataset_from_directory(data_dir,
validation_split=0.2,
    subset="training", seed=123, image_size=(224, 224), batch_size=16)
...
base_model = applications.MobileNetV2(input_shape=(224, 224, 3),
                                       include_top=False, weights="imagenet")
base_model.trainable = False
x = data_augmentation(inputs)
x = applications.mobilenet_v2.preprocess_input(x)
x = base_model(x, training=False)
x = layers.GlobalAveragePooling2D()(x)
x = layers.Dropout(0.3)(x)
outputs = layers.Dense(1, activation="sigmoid")(x)
model.compile(optimizer=keras.optimizers.Adam(1e-3), loss="binary_crossentropy",
              metrics=["accuracy"])
history_head = model.fit(train_ds, validation_data=val_ds, epochs=15)

base_model.trainable = True
for layer in base_model.layers[:100]:
    layer.trainable = False
model.compile(optimizer=keras.optimizers.RMSprop(1e-5), ...)
early_stopping = keras.callbacks.EarlyStopping(monitor="val_accuracy",
                                               patience=5,
restore_best_weights=True)
history_fine = model.fit(train_ds, validation_data=val_ds, epochs=35,
                         initial_epoch=history_head.epoch[-1],
callbacks=[early_stopping])
```

Figure 5.4: Two-phase transfer learning (excerpt from train_model.py)

5.2.5 Inference class and Base64 decoding

The FacialStressKeras class loads the trained model once at start-up and exposes a predict method that accepts a PIL image, a file path, raw bytes or a Base64 string. Figure 5.5 shows the Base64 decoding, which accepts strings with or without a data-URL prefix and repairs missing padding, a common problem with strings produced by mobile applications. If the model file is missing, the class returns a

waiting_for_model_weights status instead of failing, and any exception is returned as an error status.

```
def decode_base64_image(self, b64_str: str) -> Image.Image:
    if "," in b64_str:
        b64_str = b64_str.split(",", 1)[1]          # drop
""
    b64_str = b64_str.strip()
    missing_padding = len(b64_str) % 4
    if missing_padding:
        b64_str += "=" * (4 - missing_padding)
    image_data = base64.b64decode(b64_str)
    return Image.open(io.BytesIO(image_data)).convert("RGB")
...
score = float(self.model.predict(input_tensor, verbose=0)[0][0])
is_stressed = score > 0.5
confidence = score if is_stressed else 1.0 - score
```

Figure 5.5: Base64 decoding and classification decision (excerpt from facial_keras.py)

5.2.6 Flask endpoint

The endpoint POST /api/facial-stress, shown in Figure 5.6, accepts a JSON body with the image under image_base64 or image and an optional device_id, runs the detector, broadcasts the result to connected dashboard clients and writes it to Firebase at facial_stress/result.

```
@app.route("/api/facial-stress", methods=["POST"])
def analyze_facial_stress():
    body = request.get_json(silent=True) or {}
    image_data = body.get("image_base64") or body.get("image")
    device_id = body.get("device_id", "HEALTH_GLOVE_01")
    result = backend.facial_stress_detector.predict(image_data)
    result["device_id"] = device_id
    broadcast_event("facial_stress", result)
    if firebase_bridge:
        firebase_bridge.get_reference("facial_stress/result").set(result)
    return jsonify(result), 200
```

Figure 5.6: Facial stress endpoint (excerpt from app.py)

5.3 Integration with the Other Components

5.3.1 Glove and hub

The GSR value travels in the glove's single telemetry packet together with the heart rate, temperature and the signals used by other members' engines, under the device identifier HEALTH_GLOVE_01. The glove finds the hub by a User Datagram Protocol (UDP) broadcast, then by reading the hub address from Firebase, and finally by a fixed fallback address, and it reads the syncMode setting from the hub's /api/settings endpoint. The stress engine therefore depends on the shared glove firmware and on the hub's telemetry handling, but needs no separate connection of its own.

5.3.2 Dashboard

According to the project documents, the hub's web dashboard is rendered with Flask templates and receives live updates by Server-Sent Events (SSE). The stress estimate is presented there as a likelihood percentage, as shown in Figure 5.7.

Figure 5.7: Web dashboard showing the stress estimate (Original in Color)

5.3.3 Firebase and the mobile application

The facial exchange through Firebase is shown in Figure 5.8, and the result screen of the mobile application in Figure 5.9. The mobile application writes the request with the Base64 image, a pending flag and the device identifier; the hub's listener processes the request and writes the result, containing the detected class, the confidence and the bounding box; and the application displays it. According to the project's application integration guide, the hub also publishes the stress screening as screenings.stress, with a likelihood percentage, under /processed/HEALTH_GLOVE_01 in Firebase.

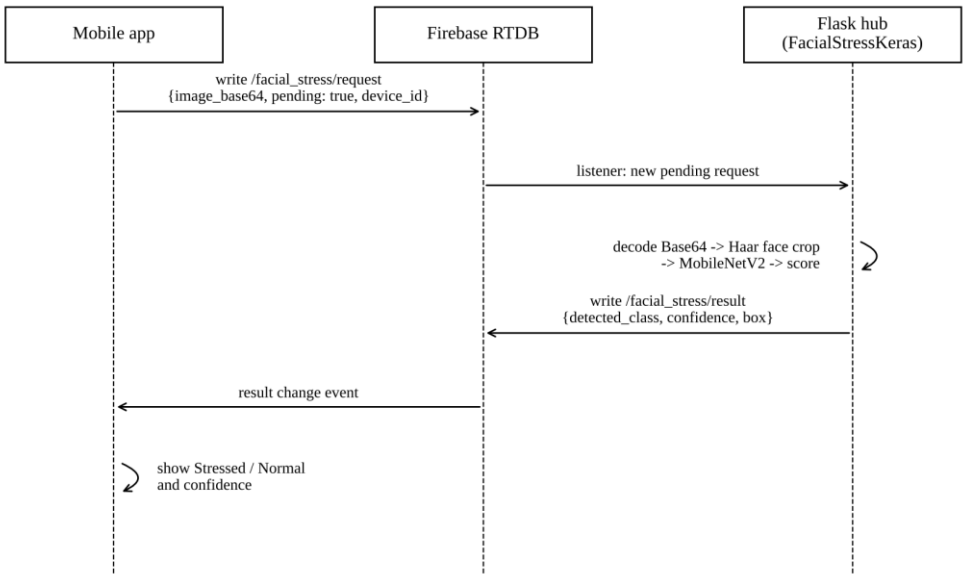


Figure 5.8: Sequence of a facial stress request through Firebase

Figure 5.9: Mobile application facial stress result screen (Original in Color)

5.3.4 Master Risk Score and alerts

A code excerpt from another member's documentation shows that, when no emergency is active, the hub computes the Master Risk Score as 0.7 times the highest likelihood

among the screening engines that are ready plus 0.3 times a correlation score. If the stress engine is one of these screening engines, a high stress likelihood would raise the Master Risk Score and could contribute to an alert, because an emergency alert is sent when the score exceeds 60 (Section 1.1). The facial result is not part of this calculation.

5.4 Verification and Software Testing

5.4.1 Model verification

For Part A, the metrics file was checked for internal consistency: the accuracy, balanced accuracy, per-class precision, recall and F1-scores recomputed from the confusion matrix matched the reported values (Section 6.1). For Part B, the saved model file was loaded with Keras 3.15.1 to confirm its structure: the augmentation block, the MobileNetV2 base with 154 layers of which 54 were trainable, the pooling, dropout and sigmoid layers, and the RMSprop optimiser with a learning rate of 1e-5, all matching train_model.py.

5.4.2 Unit and functional tests

Table 5.2 lists the tests available in the project for the component and their status.

Table 5.2: Tests of the Stress Monitoring System

ID	Test	Expected result	Observed
T1	test_gsr.ino with electrodes off	State "electrodes detached" (filtered ADC < 500)	
T2	test_gsr.ino at rest	State "normal / relaxed"	
T3	test_gsr.ino during a stressor	Rise in filtered value; possibly "high conductance / stress"	
T4	test_stress.py, 8 hand-crafted feature vectors	Relaxed cases class 0, stressed cases class 1	
T5	POST /api/facial-stress with an image	HTTP 200 with detected_class, confidence, is_stressed	
T6	POST /api/facial-stress without an image	HTTP 400 with an error message	
T7	Image without a face	Status no_face_detected	Observed (1 of 49 test crops, Section 6.2)
T8	Invalid Base64 string	Status error, is_stressed false	
T9	Firebase request-result round trip	Result written to /facial_stress/result and shown in the app	

The project's pytest file for the facial endpoint (test_facial_yolo.py) was written for the earlier YOLOv8 detector class and imports FacialStressYOLO; it does not test the FacialStressKeras class described here and needs to be adapted before it can verify the deployed model.

5.5 Deployment and Operation

The facial model file is placed in the hub's facemodel folder, where FacialStressKeras loads it at start-up; if it is absent the detector reports that it is waiting for model weights. The stress model bundle is loaded by the hub's stress engine. In operation, the user wears the glove with a finger on the pulse sensor and the GSR electrodes on the index and middle fingers; the stress estimate appears on the dashboard after the warm-up period. A facial check is started from the mobile application by taking a photograph, and the result is returned within the processing time reported in Section 6.2 plus the Firebase transfer time.

6 RESULTS AND VALIDATION

This chapter presents the results of both parts of the component. Only values that were computed by the project's code or measured for this thesis are reported; values that could not be verified are marked as missing.

6.1 Research Results

6.1.1 Part A: vital-sign stress model

Table 6.1 reports the performance of the RandomForest stress model on the 2,280 windows from the 38 held-out subjects, taken from stress_model.metrics.json, and Figure 6.1 shows the confusion matrix. All values were checked by recomputing them from the confusion matrix.

Table 6.1: Performance of the vital-sign stress model on held-out subjects

Metric	Not stressed	Stressed	Overall
Precision	98.31%	92.15%	Macro 95.23%
Recall	97.08%	95.36%	Macro 96.22%
F1-score	97.69%	93.72%	Macro 95.71%; weighted 96.64%
Support (windows)	1,677	603	2,280
Accuracy	–	–	96.62%
Balanced accuracy	–	–	96.22%

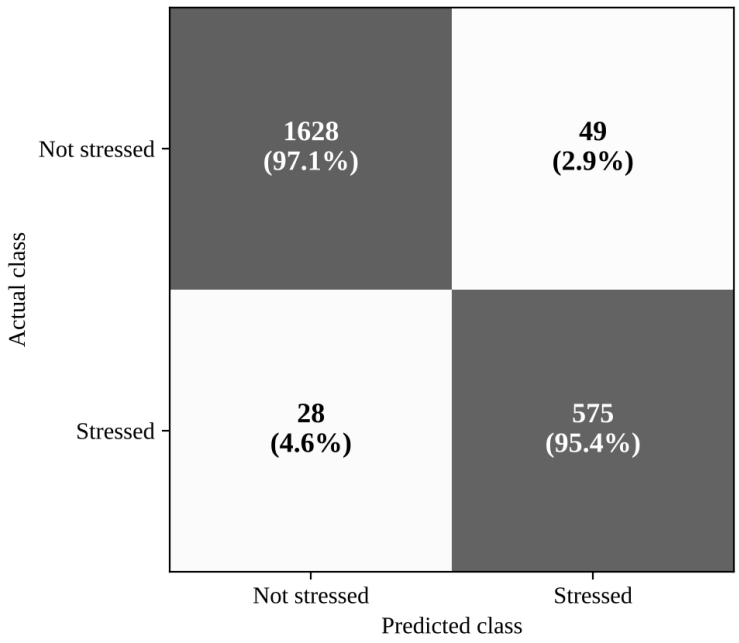


Figure 6.1: Confusion matrix of the vital-sign stress model on 2,280 test windows from 38 subjects

Of the 603 stressed windows, 575 were detected and 28 were missed, a recall of 95.36%. Of the 1,677 not-stressed windows, 1,628 were correctly classified and 49 were falsely classified as stressed, a specificity of 97.08% and a false positive rate of 2.92%. The precision of the stressed class, 92.15%, is lower than its recall because the 49 false positives are compared with a smaller stressed class. Figure 6.2 compares the precision, recall and F1-score of the two classes.

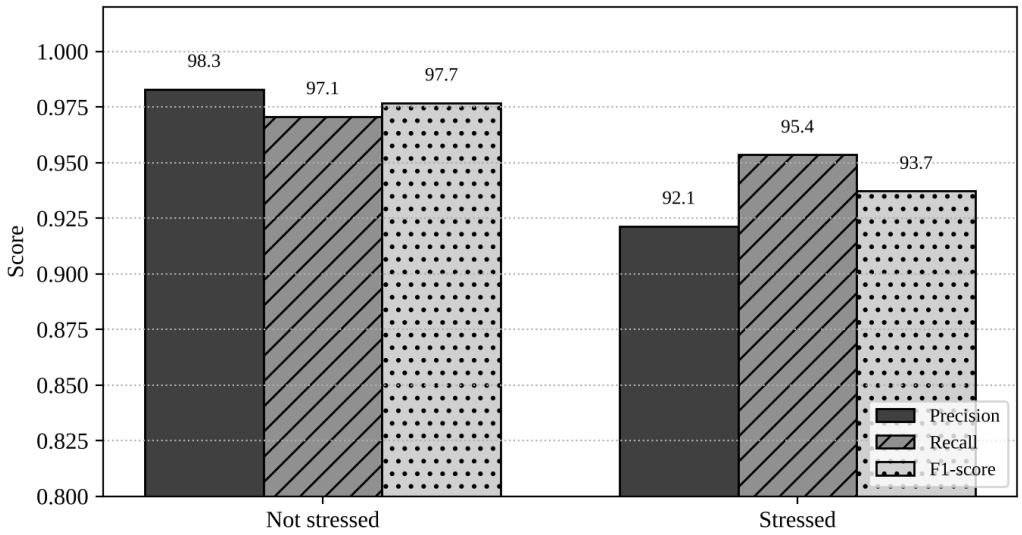


Figure 6.2: Precision, recall and F1-score of the vital-sign stress model by class

These results describe the vital-only model, P_model. The GSR blend in Equation 4.2 is not part of the trained model and was not included in this evaluation, so no claim can be made about whether GSR improves or reduces the accuracy.

6.1.2 Part B: facial stress model

Figure 6.3 shows the training and validation accuracy and loss of the facial model over both training phases; the red dashed line marks the start of fine-tuning at epoch 14. Because the validation set contains 37 images, every validation accuracy is a multiple of 1/37 (about 2.7 percentage points); one more or one fewer correct image changes the accuracy by this amount.



Figure 6.3: Training and validation accuracy and loss of the facial stress model (Original in Color)

During the first phase (epochs 0-14), with the base frozen, validation accuracy rose quickly from about 0.41 to about 0.78 by epoch 2 and reached its highest value, 0.8649 (32 of 37 images), at epochs 8 to 11. Validation loss fell steadily from about 0.76 to about 0.44, while training accuracy rose more slowly to about 0.82-0.87, which is expected because augmentation and dropout are active only during training.

In the second phase (fine-tuning), training ran for six epochs (plotted as epochs 15 to 20) before early stopping ended it. Validation accuracy was 0.8649 at epoch 15, varied between 0.81 and 0.86, and was 0.8108 (30 of 37) at the last epoch. Fine-tuning therefore did not raise the peak validation accuracy above the value already reached by the head alone. At the start of fine-tuning, training loss rose sharply from about 0.38 to about 0.54 and training accuracy fell, which indicates that unfreezing the upper base layers initially disturbed the learned features; training loss then fell to about 0.36 by epoch 20. Validation loss stayed between about 0.43 and 0.46: it reached its lowest value of about 0.43 at epoch 18 and then rose to about 0.46 at epoch 20 while training loss kept falling. This divergence is an early sign of overfitting, so the curves do not support the claim in a project summary that the model was trained "without overfitting".

Because early stopping monitored validation accuracy with restore_best_weights, the saved model has the weights of the first fine-tuning epoch at which the best fine-tuning validation accuracy occurred, and the final evaluation in train_model.py reports 86.49% on the same 37 images. This value is the peak validation accuracy, selected on

the same validation set that is used to report it, so it is optimistically biased; there is no separate test set. Table 6.2 summarises the facial results.

Table 6.2: Results of the facial stress model

Metric	Value	Status
Validation accuracy (restored best weights)	86.49% (32/37)	Verified from curves and training code
Validation accuracy at last epoch	81.08% (30/37)	Read from curves
Validation loss at restored epoch	about 0.44	Read from curves
Precision (stressed class)		Run evaluation script
Recall (stressed class)		Run evaluation script
Macro F1-score		Run evaluation script
Confusion matrix		Run evaluation script

6.2 Product and Technical Results

Table 6.3 reports the technical measurements. The facial model figures were measured for this thesis on a 2-vCPU Intel Xeon processor at 2.10 GHz with TensorFlow 2.21 and Keras 3.15.1, using the trained model file; the hub computer may be faster or slower. The end-to-end time covers Base64 decoding, Haar cascade detection, cropping and prediction with the FacialStressKeras class, measured on 49 face crops taken from the dataset folder, in 48 of which a face was detected.

Table 6.3: Technical results

Measure	Value	Source
Facial model size	16.3 MB (.keras file)	Measured
Facial model parameters	2,259,265 total; 1,862,721 trainable in the fine-tuning configuration	Measured
Facial model inference (model.predict, one image)	Median 90.1 ms; mean 93.8 ms; 95th percentile 133.6 ms (n = 50)	Measured
Facial end-to-end processing	Median 148.7 ms; mean 175.9 ms; 95th percentile 240.8 ms (n = 48)	Measured
Images with no face detected	1 of 49	Measured
Stress engine inference	About 2.9 ms per prediction	Project engineering guide
Stress engine warm-up	10 samples	Project engineering guide
GSR sampling	50 Hz, 20-sample moving average	glove.ino

Both parts are well within the near-real-time requirement NFR2. The facial value of about 85 ms stated in a project summary was typed into a document generator rather than measured; the measured median of 90 ms for the model alone is of a similar order, but the end-to-end time including face detection is about 150 ms.

6.3 User or Stakeholder Validation

No validation with older adults, carers or clinicians was carried out within the time available for this project. The usefulness and acceptability of the stress outputs to the intended users therefore remain to be established, and this is recorded as a limitation in Section 7.5 and as future work in Section 9.4.

6.4 Integration Results

The integration of the component was designed as described in Section 5.3: GSR travels in the glove's telemetry packet to the hub, the stress estimate is displayed on the dashboard and published to Firebase, and facial requests and results are exchanged through Firebase with the mobile application.

Table 6.4: Integration checks

Integration path	Result
Glove -> hub: gsr_raw received with heart rate and temperature	
Hub -> dashboard: stress likelihood displayed	
Hub -> Firebase: screenings.stress published	
App -> Firebase -> hub -> Firebase -> app: facial result displayed	
Stress likelihood reflected in the Master Risk Score	

6.5 Summary of Findings

On RQ1, the vital-only RandomForest model detected stress in windows from 38 subjects not used for training, with an accuracy of 96.62%, a balanced accuracy of 96.22% and a recall of 95.36% for the stressed class. These results depend on a dataset whose origin and labelling still need to be documented. On RQ2, the fine-tuned MobileNetV2 model reached a best validation accuracy of 86.49% (32 of 37 images), with signs of overfitting during fine-tuning and no independent test set, so its performance on new people is uncertain. On RQ3, both parts run fast enough for near-real-time use (about 3 ms for the stress engine as documented, and about 150 ms end to end for the facial pipeline as measured), and the integration paths were designed and implemented.

7 DISCUSSION

This chapter interprets the results, compares them with the literature and existing solutions, states the research and product contributions, discusses the limitations and threats to validity, and considers the implications.

7.1 Interpretation of Findings

The high balanced accuracy shows that, in this dataset, windowed statistics of heart rate and finger temperature separate stressed from not-stressed periods well. That fits the physiology behind the features: stress raises heart rate and reduces HRV [12], and it lowers peripheral temperature through vasoconstriction [2]. The model makes slightly more false-positive errors (49) than false-negative ones (28). For a screening tool whose outputs a person will review, that is the safer direction to err in. Because the split was by subject, the result is not inflated by the same person appearing in both training and test sets [7]. Still, the result is only as meaningful as the dataset. If the windows were simulated from rules or generated synthetically, the model may have learned those rules rather than real physiology, and its accuracy on real glove data could be much lower.

The facial model shows that transfer learning can reach a useful accuracy from very few images, which is what the literature leads us to expect [15], [16]. Most of the learning happened while the head was being trained; fine-tuning did not raise the peak validation accuracy and began to overfit. With only 37 validation images, the gap between 81% and 86% is two images, so the reported value is very uncertain.

7.2 Comparison with Literature Baselines and Existing Solutions

Direct comparison with published work is limited because datasets, signals, labels and evaluation protocols differ [8]. Table 7.1 places the results beside related studies.

Table 7.1: Comparison with related work

Work	Signals	Evaluation	Reported result
Schmidt et al. [5]	Wearable multimodal (WESAD)	Leave-one-subject-out	
Gjoreski et al. [9]	Wrist BVP, EDA, temperature	Laboratory and real life	
Zhang et al. [6]	Facial video	Collected dataset	

Work	Signals	Evaluation	Reported result
This work, Part A	HR and temperature windows	Subject-wise split (38 test subjects)	Accuracy 96.62%, balanced accuracy 96.22%
This work, Part B	Single face image	Hold-out of 37 images	Best validation accuracy 86.49%

Part A's accuracy is high compared with the generalisation problems reported in reviews of wearable stress detection [8], which is one more reason to establish where the dataset came from before drawing conclusions. Compared with consumer products such as smartwatch EDA scanning [19], the component measures GSR continuously instead of on demand, combines it with other vital signs, uses an open method, and ties into the health alerts of the group's platform. Unlike those products, however, it has not been validated on its intended users.

7.3 Research Contribution

The research contribution of this work is the design and evaluation of a two-part stress screening approach for older adults, combining a subject-independently evaluated vital-sign classifier with a GSR fall-back blend and a transfer-learned facial classifier, together with a critical analysis of the evaluation, including the effect of a very small validation set and of model selection on that set.

7.4 Product Contribution

The product contribution is a working Stress Monitoring System within the group's platform: GSR acquisition in the glove firmware, a stress engine on the hub that continues to work when GSR is unavailable, a facial stress detector with robust Base64 handling, an HTTP endpoint and a Firebase exchange with the mobile application.

7.5 Limitations and Threats to Validity

7.5.1 Part A: wearable stress engine

The biggest threat to the validity of the 96.62% result is that the source and labelling of the 150-subject dataset are not documented. It is not known whether the data are real, public or synthetic, or how stress was defined. Several other points also limit the work. The metrics file states a calibration to the MAX30102 sensor, whereas the glove uses the MAX30100, so the training data may not match the deployed sensor. The test script describes a 60 s window while the design states 120 s. The GSR score is a fixed-

weight heuristic outside the trained model, with fixed ADC bounds of 300 and 3200 and no per-user baseline, even though EDA varies greatly between people [11]. Its weight of 0.2 was not validated, and the reported accuracy describes the vital-only model. GSR is reported in raw ADC counts rather than microsiemens, and the ESP32 ADC is not calibrated. Heart rate and finger temperature are sensitive to movement, posture, illness and room temperature, and the engine does not use activity context. Telemetry is sent only when a finger is on the pulse sensor, and data recorded with syncMode on contain simulated heart rate and temperature, so they are invalid for stress evaluation. Finally, no clinical validation was done and no older adults took part.

7.5.2 Part B: facial model

The facial dataset is small (189 images) and the validation set has only 37, so each image moves the accuracy by about 2.7 percentage points. The images seem to come from few people, and since the split is by image rather than by person, the same person probably appears in both training and validation sets, which inflates the result [7]. The expressions may be posed and may not reflect physiological stress. There is no separate test set, and the reported 86.49% is the peak validation accuracy, picked on the same set; the last epoch reached about 81%. After fine-tuning, validation accuracy fluctuated and validation loss rose slightly at the end. The people in the dataset do not appear to be older adults, whose faces and expressions differ. Precision, recall and the confusion matrix have not yet been computed. A single photograph also captures only one moment and depends on lighting and pose.

7.5.3 Integration and security

Communication between the glove and the hub is not encrypted or authenticated, credentials are compiled into the firmware, and facial images are stored temporarily in Firebase, as discussed in Section 3.6. The project documents disagree on several points (for example HTTP versus Message Queuing Telemetry Transport (MQTT) transport, MobileNetV2 versus YOLOv8, and the dataset sizes in a draft checklist), which shows the need for a single, verified technical record.

7.6 Implications

In practice, the results suggest that the sensors already on a health monitoring glove can support stress screening at almost no extra cost, and that a phone camera offers a complementary, contact-free check. Before the outputs are used to inform care, though, both models need evaluating on real data from older adults, with documented labels and consent. For research, the work shows how easily small validation sets and selection on the validation set inflate reported accuracy, and why subject-wise evaluation and independent test sets matter [7], [8].

8 COMMERCIALISATION AND SUSTAINABILITY

This chapter presents an analysis of the commercial and sustainability aspects of the Stress Monitoring System. It is an analysis of potential, not a statement of facts about a market; no market sizes or prices are stated without a source.

8.1 Value Proposition and Target Users

The primary users are older adults living at home or in care homes; the secondary users are family carers, care home staff and healthcare providers who receive the information. The value proposition is continuous, non-invasive stress screening that uses sensors already worn for other health purposes, an optional contact-free facial check from a phone, and the presentation of stress information alongside other health indicators in one dashboard and application, so that carers can notice changes in well-being early.

8.2 Market and Competitive Position

Competing approaches include consumer smartwatches with stress features, such as EDA scanning [19], and research wristbands [9]. Table 8.1 compares them with the component.

Table 8.1: Competitive comparison (analysis)

Feature	Consumer smartwatch	Research wristband	Stress Monitoring System
Target users	Healthy adults	Researchers	Older adults and carers
Stress signals	HR/HRV; EDA on demand	BVP, EDA, temperature	HR, finger temperature, continuous GSR; face photo
Integrated with other elderly health alerts	No	No	Yes
Open, explainable method	No	Partly	Yes
Clinically validated	Varies	Research use	No (prototype)
Additional hardware cost	Device price	Device price	Low (GSR module on the existing glove; no price stated)

8.3 Business or Sustainability Model

The component is not intended to be sold separately; its value lies in adding stress screening to the group's monitoring kit. A possible model is a one-time hardware price for the glove and belt with an optional subscription for the cloud dashboard and application, offered to families and care homes. The added cost of the stress

component is mainly the GSR module and electrodes, since the facial check uses the user's existing phone and the hub's existing computer.

8.4 Scalability Intellectual Property and Compliance

Each glove identifies itself by a device identifier, so additional users can be added on the hub; however, the facial result is currently written to a single Firebase path, which would need to be separated per user for multi-user deployment. The models are small enough to run on an ordinary computer, and the RandomForest and MobileNetV2 inference times allow many users per hub.

The methods used (random forests, MobileNetV2, Haar cascades) are published and implemented in open-source libraries whose licences must be respected; the value of the component lies in its integration and data rather than in patentable methods. Facial and physiological data are special categories of personal data, so compliance with Sri Lanka's Personal Data Protection Act [20], including consent, purpose limitation, security and deletion of images after processing, is a precondition for deployment. A product that made health claims could also fall under medical device regulation, in which case IEC 62304 [21] and ISO 14971 [22] would apply.

8.5 Contribution to the Sustainable Development Goals

The component contributes mainly to Sustainable Development Goal (SDG) 3, Good Health and Well-being, which includes the promotion of mental health and well-being [23], by helping carers to notice stress in older adults. By using low-cost sensors and an ordinary phone, it also supports SDG 10, Reduced Inequalities, in the sense of making monitoring technology more affordable; this is an analysis of potential rather than a measured impact.

9 CONCLUSION AND FUTURE DIRECTIONS

9.1 Conclusion

This thesis presented the Stress Monitoring System, my component of a multi-sensor wearable platform for older adults. Part A reads GSR on the glove at 50 Hz with a moving-average filter and estimates stress on the hub with a RandomForest on windowed heart-rate and temperature statistics. It blends in a GSR score and falls back to the vital-only model when the electrodes are detached. Evaluated on 38 subjects not used for training, the vital-only model reached an accuracy of 96.62% and a balanced accuracy of 96.22%. Part B classifies facial photographs from the mobile application with a Haar cascade and a fine-tuned MobileNetV2, reaching a best validation accuracy of 86.49% on 37 images and an end-to-end processing time of about 150 ms on a CPU. Low-cost stress screening therefore looks technically feasible within the platform. Its real-world performance, however, has not yet been shown, because the origin of the physiological dataset is undocumented, the facial dataset is small, and nothing has been evaluated on older adults.

9.2 Achievement of Objectives

Table 9.1 maps each specific objective in Section 1.5.2 to its status and evidence.

Table 9.1: Achievement of objectives

Objective	Status	Evidence
1. GSR acquisition and transmission	Achieved	Section 5.2.1, Figure 5.1
2. Subject-independent stress classifier	Achieved (dataset provenance to be documented)	Table 6.1, Figure 6.1
3. GSR blend with fall-back	Partly achieved (implemented, not validated)	Section 4.4.2; no with/without-GSR experiment
4. Facial stress classifier	Partly achieved	Figure 6.3, Table 6.2; precision, recall and test set missing
5. Integration with hub, Firebase, dashboard and app	Partly achieved (implemented; integration results to be recorded)	Section 5.3, Table 6.4
6. Latency and functional verification	Partly achieved	Table 6.3 (latency measured); Table 5.2 (tests to be reported)

9.3 Individual Research and Product Contributions

Research contributions. This thesis contributes the design and evaluation of a two-part stress screening approach for older adults: a vital-sign classifier evaluated subject-

wise on 38 unseen subjects (accuracy 96.62%, balanced accuracy 96.22%) with a GSR fall-back blend, and a transfer-learned facial classifier. It also provides a critical analysis of the evaluation, in particular the effect of a very small validation set and of selecting the best epoch on that set, and it records the dataset provenance, labelling and validation gaps that must be closed before the models are used with older adults.

Product contributions. The product contribution is a working Stress Monitoring System within the group's platform: GSR acquisition in the glove firmware, a hub-side stress engine that continues to work when GSR is unavailable, a facial stress detector with robust Base64 handling, an HTTP endpoint, and a Firebase exchange with the mobile application.

9.4 Future Directions

Six pieces of future work follow from these findings. First, the physiological model should be retrained and evaluated on real data recorded with the glove, or on a documented public dataset such as WESAD [5], using subject-wise validation. Second, GSR should be converted to conductance, normalised to a per-user baseline and included as features in the trained model, with an experiment comparing performance with and without it. Third, activity context from the belt could help separate stress from physical exertion, as the original system overview and the literature [9] suggest. Fourth, the facial dataset should be enlarged with consenting participants, including older adults, and split by person into training, validation and test sets, with precision, recall and confusion matrices reported on the test set. Fifth, the two parts could be combined into one multimodal estimate. Finally, communication should be secured, images deleted after processing and Firebase access restricted, and the system should be evaluated with older adults, carers and clinicians for usefulness and acceptability.

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GLOSSARY

Balanced accuracy – The mean of the recall of each class; it is not inflated by a majority class.

Electrodermal activity – Changes in the electrical properties of the skin caused by sweat gland activity; measured as galvanic skin response or skin conductance.

Fine-tuning – Further training of some layers of a pre-trained network with a low learning rate.

Haar cascade – A fast face detector built from simple rectangular features arranged in a cascade of classifiers.

Moving average – A filter that replaces each value with the mean of the most recent samples.

Subject-wise split – A division of data in which all samples from one person are placed in either the training set or the test set, never both.

syncMode – A glove setting that replaces measured heart rate, SpO2 and temperature with simulated healthy values.

Transfer learning – Reusing a model trained on one task as the starting point for another task.

APPENDICES

Appendix A Extended Results and Source Listings

A.1 Metrics file of the vital-sign stress model

Figure A.1 lists the complete metrics file produced when the stress model was evaluated on the held-out subjects (Section 6.1.1).


```
{
  "accuracy": 0.9662280701754385,
  "balanced_accuracy": 0.9621733313159901,
  "classification_report": {
    "not_stressed": {
      "precision": 0.9830917874396136,
      "recall": 0.9707811568276684,
      "f1-score": 0.976897689768977,
      "support": 1677.0
    },
    "stressed": {
      "precision": 0.9214743589743589,
      "recall": 0.9535655058043118,
      "f1-score": 0.9372453137734311,
      "support": 603.0
    },
    "accuracy": 0.9662280701754385,
    "macro avg": {
      "precision": 0.9522830732069862,
      "recall": 0.9621733313159901,
      "f1-score": 0.957071501771204,
      "support": 2280.0
    },
    "weighted avg": {
      "precision": 0.9667955991218292,
      "recall": 0.9662280701754385,
      "f1-score": 0.9664106798017339,
      "support": 2280.0
    }
  },
  "confusion_matrix": [
    [
      1628,
      49
    ],
    [
      28,
      575
    ]
  ],
  "train_windows": 6720,
  "test_windows": 2280,
  "train_subjects": 112,
  "test_subjects": 38,
  "sensor_calibration": "MAX30102 + MLX90614 (SmartGlove)"
}
```


Figure A.1: stress_model.metrics.json

A.2 GSR test sketch

Figure A.2 lists the standalone GSR test sketch (test_gsr.ino) used to check the sensor (Section 4.4.1).

```
/*
 * ============================================================================================
 *  GSR (Galvanic Skin Response / Skin Conductance) Standalone Test Sketch
 *  Target Board: ESP32 Dev Module (WROOM-32) / ESP32-S3
 * ============================================================================================
 *
 *  WIRING (GSR Sensor <-> ESP32):
```

```
 *  ------------------------------------------------------------
 *  GSR Sensor Pin       ESP32 DevKit (WROOM)       ESP32-S3 Mini
 *  --------------       --------------------       --------------
 *  VCC                  3.3V (or 5V)               3.3V
 *  GND                  GND                        GND
 *  SIG / OUT (Analog)   GPIO 34 (ADC1_CH6)         GPIO 1
 *  ------------------------------------------------------------
 *
 *  CALIBRATION / CONDUCTANCE:
 *    - Resistance R (ohms) decreases as skin sweat / stress increases.
 *    - Conductance (microsiemens, uS) increases when stressed.
 * ============================================================================================
 */

// Choose pin according to board:
// ESP32 DevKit: GPIO 34 (ADC1_CH6 - WiFi safe)
// ESP32-S3 Mini: GPIO 1
constexpr int GSR_PIN = 34;

// Rolling average filter for 50Hz mains noise rejection
constexpr int NUM_READINGS = 20;
int readings[NUM_READINGS];
int readIndex = 0;
long total = 0;
int averageGSR = 1;

unsigned long lastSampleTime = 0;
unsigned long lastPrintTime = 0;

void setup() {
  Serial.begin(115200);
  while (!Serial) {
    delay(10);
  }

  Serial.println("\n=====================================================");
  Serial.println("        GSR (Skin Conductance) Sensor Test");
  Serial.println("=====================================================");
  Serial.printf("Configured GSR Analog Pin: GPIO %d\n", GSR_PIN);

  // Configure ESP32 12-bit ADC (0 - 4095)
  analogReadResolution(12);
  analogSetAttenuation(ADC_11db); // Full range ~0V to 3.3V
  pinMode(GSR_PIN, INPUT);

  // Initialize smoothing buffer
  int initialRead = analogRead(GSR_PIN);
  for (int i = 0; i < NUM_READINGS; i++) {
    readings[i] = initialRead;
    total += initialRead;
  }
  averageGSR = initialRead;

  Serial.println("Place electrodes on two fingers (index & middle fingers)...\n");
}

void loop() {
  unsigned long now = millis();

  // Sample at 50 Hz (every 20 ms)
  if (now - lastSampleTime >= 20) {
    lastSampleTime = now;

    int rawGSR = analogRead(GSR_PIN);

    // Moving average filter
    total = total - readings[readIndex];
    readings[readIndex] = rawGSR;
    total = total + readings[readIndex];
    readIndex = (readIndex + 1) % NUM_READINGS;
    averageGSR = total / NUM_READINGS;
  }

  // Print output every 250 ms (4 Hz)
  if (now - lastPrintTime >= 250) {
    lastPrintTime = now;
```

```
    float voltage = (averageGSR * 3.3f) / 4095.0f;

    // Estimate skin resistance (assuming standard 100k voltage divider)
    float resistanceK = 0.0;
    if (voltage > 0.05 && voltage < 3.25) {
      resistanceK = (3.3f - voltage) / (voltage / 100.0f);
    }

    // Determine state
    String state = "IDLE";
    if (averageGSR < 500) {
      state = "ELECTRODES DETACHED";
    } else if (averageGSR > 2800) {
      state = "HIGH CONDUCTANCE / STRESS";
    } else {
      state = "NORMAL / RELAXED";
    }

    Serial.printf("[GSR] Raw: %4d | Filtered: %4d | Voltage: %4.2f V | R: %6.1f kOhm |
Status: %s\n",
                  readings[(readIndex - 1 + NUM_READINGS) % NUM_READINGS],
                  averageGSR, voltage, resistanceK, state.c_str());
  }
}
```

Appendix B Hardware Connections and System Configuration

This appendix summarises how the Stress Monitoring System is connected and configured. The block diagram of the component and its interfaces is Figure 4.1, the GSR signal chain is Figure 4.2, and the sensor front end is described in Section 4.4.1. The values below are those recorded in the project firmware and in Table 3.2.

Table B.1: GSR module wiring

GSR module pin	ESP32 DevKit (WROOM-32)	ESP32-S3 Mini
VCC	3.3 V (or 5 V)	3.3 V
GND	GND	GND
SIG / OUT (analog)	GPIO 34 (ADC1_CH6)	GPIO 1

Table B.2: Data path of the Stress Monitoring System

Stage	Where it runs	Description
GSR acquisition	Glove node (ESP32)	GSR read on GPIO 34 (12-bit, 11 dB attenuation), sampled every 20 ms (50 Hz) and smoothed with a 20-sample moving average; sent as gsr_raw in ADC counts.
Transmission	Glove node to hub	One JSON packet per second, sent by HTTP POST over Wi-Fi to the /api/telemetry endpoint while a finger is detected on the pulse sensor.
Vital-sign stress engine	Hub (Flask backend)	Rolling 120 s buffer; mean, standard deviation, minimum and maximum of heart rate and temperature; RandomForest (300 trees); result blended with the GSR score as P_final = 0.8 P_model + 0.2 S_gsr, with S_gsr taken from ADC values 300 to 3200.
Facial stress pipeline	Mobile app, Firebase, hub	Photograph from the mobile application passed through Firebase as Base64, face found with a Haar cascade, 224 × 224 crop with 10% padding classified by the fine-tuned MobileNetV2 through the /api/facial-stress endpoint.
Output	Hub, dashboard, mobile app	Stress estimates shown on the web dashboard and in the mobile application as non-diagnostic screening information.